# MEMORANDUM OF AGREEMENT

KNOW ALL MEN BY THESE PRESENTS:

This Memorandum of Agreement is made and entered into by and among:

The **Department of Environment and Natural Resources – Regional Office V**, with principal Office at Rawis, Legazpi City, herein represented by its Regional Director, **CRISANTA MARLENE P. RODRIGUEZ**, hereinafter referred to as "DENR ROV";

The **Partido State University**, with Principal Office at Goa, Camarines Sur represented by its President, **RAUL G. BRADECINA**, hereinafter referred to as the "PSU".; and

The **Department of Agriculture-Bureau of Fisheries and Aquatic Resources – Regional Office V**, with principal Office at Bula, Camarines Sur herein represented by its Regional Director, **DENNIS V. DEL SOCORRO**, hereinafter referred to as "DA-BFAR ROV";

WITNESSETH:

**WHEREAS,** the DENR is the primary government agency responsible for the conservation, management, development and proper use of the country's environment and natural resources, specifically forest and grazing lands, mineral lands, including those in reservation and watershed areas, and lands of the public domain, as well as the licensing and regulation of all natural resources as may be provided for by law in order to ensure equitable sharing of the benefits derived therefrom for the welfare of the present and future generations of Filipinos;

**WHEREAS**, Executive Order No. 533 was promulgated and otherwise known as: "Adopting the Integrated Coastal Management (ICM) as a National Strategy to Ensure the Sustainable Development of the Country's Coastal and Marine Environment and Resources and Establishing Supporting Mechanisms for its Implementation";

**WHEREAS**, DENR Department Administrative Order No. 2016-26 was issued to set the Guidelines for the implementation of the Coastal and Marine Ecosystem Management Program (CMEMP);

**WHEREAS**, the Malabungot Protected Landscape and Seascape (MPLS) was established through Presidential Proclamation No. 288 in 2000;

**WHEREAS**, the DENR-ROV, in order to carry-out the Coastal and Marine Ecosystem Management Program (CMEMP) in Malabungot Protected Landscape and Seascape (MPLS) will conduct an "Applied *Research on the Mapping of Coastal*

*Habitats for Potential Ecotourism Livelihood in Malabungot Protected Landscape and Seascape, Binagasbasan, Garchitorena, Camarines Sur";*

**WHEREAS**, the DA-BFAR ROV is responsible for the development, improvement, management and conservation of the country's fishery and aquatic resources;

**WHEREAS**, PSU was mandated through Republic Act 9029 to provide advanced instruction and training in the arts, philosophy, sciences, technology and other graduate and professional courses. PSU was further mandated to undertake advanced studies, research, production and extension services and shall provide leadership in these areas;

**WHEREAS**, the Partido State University (PSU) will be the Lead Academic Institution for this research project;

**NOW THEREFORE**, the Parties by mutual covenants herein set forth agree as follows:

## I. PROJECT DESCRIPTION

Applied *Research on the Mapping of Coastal Habitats for Potential Ecotourism Livelihood of Malabungot Protected Landscape and Seascape located at Garchitorena, Camarines Sur under DENR Administrative Order No. 2016-26*

## II. RESPONSIBILITIES OF THE PARTIES

### A. The DENR-ROV shall:

1. Provide financial support in the amount of Php1,300,000.00 for the Coastal and Marine Ecosystem Management Program (CMEMP) involving the conduct of the project *entitled* "Applied *Research on the Mapping of Coastal Habitats for Potential Ecotourism Livelihood in Malabungot Protected Landscape and Seascape, Binagasbasan, Garchitorena, Camarines Sur";*

2. Upon request, DENR ROV shall provide technical inputs and other support needed in the preparation of the research plan and project implementation. *DENR shall disburse the funds to cover all expenses of the research project;*

3. Ensure that specific roles and work plans among the contracting parties are adhered to and sustained, but not limited to, the budget and deliverables necessary to implement the plans;

4. In coordination with the PENR Office of Camarines Sur which shall be the field representative of DENR ROV shall supervise the efficient,

effective and orderly field operations of PSU and shall conduct regular monitoring and performance evaluation in the implementation of the project;

5. Ensure that funds are disbursed/utilized in accordance with existing accounting and auditing rules and regulations; and

6. Conduct inspection and validation of the research study.

**B. The PSU shall:**

1. In coordination with DENR Region V, prepare an inception/concept paper to ensure common understanding among parties on the plan of action and timeline of the activities to be undertaken. It shall also provide mutual interpretation of the Terms of References (TOR) of this Agreement;

2. In close coordination with DENR ROV and PENRO-Camarines Sur and BFAR, implement research project entitled, "*Malabungot Protected Landscape and Seascape, Garchitorena, Camarines Sur*";

3. Conduct hands-on capability enhancement training activities to the technical staff of DENR ROV. Likewise, PSU shall provide hands-on opportunities to develop capacities along biodiversity conservation and other ICM-related studies;

4. Conduct studies on Socio-economic and Livelihood Economic Valuation, Ecotourism Development Planning Project and Total Economic Valuation Techniques;

5. Provide technical backstopping with BFAR on Mapping of Coastal Habitats study component and total economic valuation studies;

6. Ensure that all project objectives and expected outputs of the above research studies are adequately addressed in accordance with the Work and Financial Plan and made an integral part of this agreement until December 2017;

7. Submit terminal report in a scientific technical format, outlining results and findings of the research studies entitled, "(1) *Socio-economic and livelihood Status of Malabungot Protected Landscape and Seascape Communities; (2) Economic Valuation of Malabungot Protected Landscape and Seascape Ecosystems; (3) Ecotourism Plan for Malabungot Protected Landscape and Seascape" and (4) "Assessment of Seagrass, Plankton, Soft-Bottom Communities and Reef-Associated Cryptobiota of Malabungot Protected Landscape and Seascape, Garchitorena, Camarines Sur*";

8. Present the project results in a forum, research review and/or scientific meetings for the purpose of validating the results and dissemination;

9. Conduct hands-on capability enhancement training activities to the technical staff, DENR ROV along conventional coastal habitat assessment, coastal vulnerability assessment and capture fisheries assessment as inputs for economic valuation techniques;

10. Conduct Mapping of Coastal Habitats- survey, GIS mapping and ground truthing survey using precise survey instrument to come-up with digital thematic maps with updated geographic information system (GIS) software. Among others, the following maps are required: individual maps of mangrove forest, seagrass beds, coral reefs, map of the Malabungot Protected Landscape and Seascape (MPLS);

11. Submit report in scientific format digital thematic maps using GIS of mangrove forest, seagrass, coral reefs of *Malabungot Protected Landscape and Seascape;*

12. Ensure that all project objectives and expected outputs of the above research studies are adequately addressed in accordance with the Work and Financial Plan and made an integral part of this agreement until December 2017;

13. Agree, together with DENR ROV, the work outputs in accordance with the attached work and financial plan; and

14. Regularly submit monthly status report to DENR ROV of the conduct of the study.

**C. The DA-BFAR shall:**

1. Assist the PSU in the conduct of Habitat Assessment of the research study entitled: "Applied *Research on the Mapping of Coastal Habitats for Potential Ecotourism Livelihood*" through Fish Visual Census and applicable coastal assessments inherent to the mandates of the BFAR.

## III. EFFECTIVITY AND DURATION OF THE AGREEMENT

The research study shall take effect upon execution and shall remain as such until December 31, 2017 unless a written request for extension of the project implementation be initiated by the PSU in coordination with DA-BFAR under valid justifications and written approval issued by the DENR ROV prior to the expiration of the original agreement. Implementation of the research study mentioned in this agreement shall be based on agreed work and financial plan and *shall commence upon signing of the AGREEMENT.*

## IV. PAYMENT SCHEDULE

A total amount of *One Million Three Hundred Thousand (P1,300,000.00)* shall be allotted by DENR ROV to pay the cost of the research project. It shall be released in accordance with the work and financial plan.

## V. INTELLECTUAL PROPERTY

PSU and BFAR and its researchers recognize the results of the programs/projects and the right to publication, except those that relate to intellectual property concerns, belonging jointly to them and to the DENR ROV.

Full acknowledgement shall be accorded to the DENR ROV in case the research results are published or presented in various fora, seminars and meetings. Such presentation should not prejudice the propriety/confidential nature of the information as heretofore mentioned for purposes of intellectual property protection.

However, prior approval of DENR ROV shall be solicited whenever the end product of this project will be used for profit. If such is the case, the DENR ROV shall demand as a condition, the payment of royalties. No prior approval shall be required if such is used in assemblies and/or meetings of public character.

Subject to the limitations recognized by law, particularly those that relate to intellectual property (IP), the Parties reserve the right to use all data and findings by virtue of and pursuant to this Agreement for the enhancement of its academic and research programs.

## VI. WARRANTIES

Each Party represents and warrants to the other that:

a. It is an institution duly organized, validly existing and in good standing under the laws of the jurisdiction of its institution;

b. The execution, delivery and performance of this Agreement have been duly authorized by all necessary corporate actions and will not result in any breach or violation of any of its existing charter documents, contracts, licenses, permits and authorizations;

c. This Agreement, when executed constitutes a valid and binding obligation on the Parties and shall be enforceable in accordance with its terms; and

d. All authorizations, licenses with regulations and consents from their respective charters, as required by law respect to the execution and performance of this Agreement, have been duly taken, and shall be kept valid and effective by the Parties during the term of this Agreement.

## VII. LIABILITY AND INDEMNIFICATION

PSU and BFAR ROV shall indemnify, DENR ROV, its officers, employees, agents, representatives and their respective successors or assigns from all costs, expenses, damages, loses and attorney's fees incurred and arising out of any breach or alleged breach by PSU and BFAR ROV of any terms and conditions contained in this Agreement or of any representations or warranties PSU, BFAR ROV have made herein.

Should either party incur delays or fail to comply with the expressed terms and conditions provided herein due to causes such as but not limited to *force majeure* or incidents of similar nature which are beyond the control of the party so failing, shall not be deemed to be a breach of this Agreement and shall not subject either party to liability. However, each party is solely responsible for any eventualities or accident incurred during the course of the project, if such is connected to a contract made to a third party which is not part of this Agreement.

## VIII. RELATIONSHIP OF PARTIES

This Agreement does not constitute a partnership agreement or create an employee-employer relationship, joint venture, or agency relationship between the parties. No Party shall hold itself out contrary to the terms and conditions of this Agreement. It is specifically understood that each party is an independent entity and shall not be considered an employer, employee, agent or representative of the other. Each Party and/or its employees may not bind any other party to any obligation or contract.

## IX. DISCLAIMER

The Parties to this Memorandum of Agreement shall maintain their own separate missions, mandates and their own accountabilities. Nothing in this Memorandum of Agreement shall be construed as superseding or interfering in any way with other agreements or contracts entered into between the Parties, either prior to or subsequent to the signing of this Memorandum of Agreement.

## X. MISCELLANEOUS

Any provision of this Memorandum of Agreement found to be prohibited by law shall be ineffective to the extent of such prohibition without invalidating the remainder of this Agreement. This Memorandum of Agreement represents the entire intention of the parties and may not be amended except in writing signed by each Party hereto. Each Party to this Memorandum of Agreement warrants to the other that it is duly organized and existing and each signatory hereto represents to the other party that it has full right and authority to enter into and consummate this Memorandum of Agreement and all related documents.

## XI. ETHICAL CONDUCT OF RESEARCH

PSU and BFAR has the responsibility of ensuring that the conduct of the research is in accordance with existing ethical research standards.

## XII. SEVERABILITY

If any condition or provision of this Agreement is found by any court or administrative body of competent jurisdiction to be invalid or unenforceable to any extent, such condition or provision shall be severed from the remaining conditions

and provisions, and all conditions and provisions not affected by such invalidity or unenforceability shall remain in force and effect. The Parties may agree to amend any invalid or unenforceable condition or provision to achieve to the greatest extent possible the objectives of the previously invalid or unenforceable condition or provision, provided however, that the resulting amendment is in such a manner as to leave the amended agreement substantially the same.

## XIII. DOCUMENTS COMPRISING THIS AGREEMENT

All documents hereto attached are hereby expressly made an integral part of this Agreement by reference, excluding inconsistencies with any/all part, terms, and conditions contained in this Memorandum of Agreement.

## XIV. ALTERNATIVE DISPUTE RESOLUTION

Any dispute or claim arising out of or from this Agreement shall be settled by informal dispute proceedings, the process of which shall be mutually agreed upon by the Parties. Should Parties fail to obtain resolution to a particular dispute or claim arising out of or in connection with this Agreement, such dispute or claim shall then be referred to and finally resolved by binding arbitration in accordance with Philippine laws.

## XV. GOVERNING LAW

This Agreement shall be governed by and construed in accordance with the laws of the Republic of the Philippines.

IN WITNESS WHEREOF, the Parties represented by their duly authorized officers have signed this Memorandum of Agreement at the place and on the date above written.

**DEPARTMENT OF ENVIRONMENT NATURAL RESOURCES**

**DIR. CRISANTA MARLENE P. RODRIGUEZ**
Regional Director

**PARTIDO STATE UNIVERSITY**

**RAUL G. BRADECINA, Ph.D.**
SUC President III

**DEPARTMENT OF AGRICULTURE-BUREAU OF FISHERIES AND AQUATIC RESOURCES ROV**

**DIR. DENNIS V. DEL SOCORRO**
Regional Director

**RAMON F. FUENTEBELLA**
Witness

**ENGR. ROBERTO D. SHEEN**
OIC, ASSISTANT REGIONAL DIRECTOR FOR TECHNICAL SERVICES
Witness

**MELCHOR B. DERAMAS**
*Chief, Admin & Finance Division*
Witness

## ACKNOWLEDGEMENT

REPUBLIC OF THE PHILIPPINES)

**BEFORE ME**, a Notary Public for and in the Legazpi, personally appeared in this 5th day of September 2017 at the following:

| NAME | ID. NO | ISSUED BY |
|---|---|---|
| **CRISANTA MARLENE P. RODRIGUEZ** | E02-98-036394 | LTFRB |
| **RAUL BRADEGINA, Ph.D.** | EC 6518477 | Passport |
| **DENNIS V. DEL SOCORRO** | CRN 006-0071-5938-5 | SSS |

Known to and to me known to be the same persons who executed the foregoing instrument and they acknowledged to me that same is their free and voluntary act and deed as well as that of the entities they respectfully represent.

WITNESS MY HAND AND SEAL on the date and place first above-written.

ATTY. MELISSA A. ABADEZA
NOTARY PUBLIC
FOR AND IN THE CITY OF LEGAZPI
AND [illegible]
UNTIL DEC. 31, 2017
ROLL NO. 56326
PTR NO. [illegible]-K 1-5-2016
IBP NO. 1029303 1-21-2016
MCLE COMPLIANCE NO. III-0019052

NOTARY PUBLIC

Doc. No. 1878
Page No. 60
Book No. IV
Series of 2017

# SUPPLEMENTAL MEMORANDUM OF AGREEMENT

(TO THE MEMORANDUM OF AGREEMENT ENTERED INTO BY AND AMONG THE DENR V, PARTIDO STATE UNIVERSITY AND DA-BFAR V ON JUNE 6, 2017)

**KNOW ALL MEN BY THESE PRESENTS:**

This Supplemental Memorandum of Agreement is made and entered into this _____ of _____ 2018 at the City of Legazpi, by and among:

The **Department of Environment and Natural Resources Regional Office V,** a government entity with principal office at Regional Center Site, Rawis, Legazpi City, herein represented by its Regional Director, **ATTY. ANTONIO A. ABAWAG,** CESO IV, herein referred to as "**DENR V**",

-and-

The **Department of Agriculture-Bureau of Fisheries and Aquatic Resources-Regional Office V**, with principal office at Bula, Camarines Sur, herein represented by its Regional Director, **DENNIS V. DEL SOCORRO**, hereinafter referred to as "**BFAR V**",

-and-

The **Partido State University**, with principal office at Poblacion, Goa, Camarines Sur, herein represented by its President, **RAUL G. BRADECINA**, herein referred to as the "**PSU**".

**WITNESSETH:**

**WHEREAS,** the DENR, PSU and DA-BFAR have previously entered into an Agreement on June 5, 2017, consistent with DENR Administrative Order No. 2016-26 for the implementation of the Coastal and Marine Ecosystem Management Program (CMEMP) in Malabungot Protected Landscape (MPL).

**WHEREAS,** the DENR V, BFAR and PSU agreed to converge for the implementation of the "Applied Research on Mapping of Coastal Habitats for Potential Tourism Livelihood in Malabungot Protected Landscape (MPL) in Binagasbasan, Garchitorena, Camarines Sur".

**WHEREAS,** the Partido State University (PSU), being an academic institution, took the lead in the execution of this research project;

**WHEREAS,** Section II.B.12 of the aforenoted Memorandum of Agreement stipulated to "ensure all project activities and expected outputs of the applied research studies should have been addressed consistent with the work and financial plan until December, 2017";

**WHEREAS,** the Partido State University, through the letter of President Raul G. Bradecina dated December 29, 2017 to then DENR Regional Director Crisanta Marlene P. Rodriguez requested for an extension of the applied research project due to persistent incremental bad weather condition especially during the fourth quarter of CY 2017; and

**WHEREAS,** a Memorandum dated 08 January 2018 of Asst. Regional Director Roberto D. Sheen of Technical Services indorsed the aforementioned letter which was duly approved by then Regional Director Crisanta Marlene P. Rodriguez.

**NOW THEREFORE,** the parties by mutual covenants and considerations herein set forth agreed as follows:

The previous MOA which was executed on June 5, 2017, which were represented by then DENR Regional Director Crisanta Marlene P. Rodriguez, PSU President Raul G. Bradecina and DA-BFAR Regional Director Dennis V, Del Socorro executed the same Memorandum of Agreement enter into this Supplemental Memorandum of Agreement (the "**Supplemental MOA**") to extend the previous MOA until December 2018 with the same terms and conditions. All the terms and responsibilities of the MOA remain unchanged.

**IN WITNESS WHEREOF**, the Parties represented by their duly authorized officers have signed this Memorandum of Agreement on the date above written.

DEAPRTMENT OF ENVIRONMENT AND NATURAL RESOURCES:

**ATTY. ANTONIO A. ABAWAG, CESO IV**
OIC, Regional Director

PARTIDO STATE UNIVERSITY:

**RAUL G. BRADECINA, Ph.D**
SUC President III

DEPARTMENT OF AGRICULTURE-
BUREAU OF FISHERIES AND
AQUATIC RESOURCES V:

**DIR. DENNIS V. DEL SOCORRO**
Regional Director

ALLAN L. LEUTERIO
ARD FOR TECHNICAL SERVICES
Witness

PATRICIA M. CANDELARIA
Witness

NONIE P. ENOLVA
Witness

## ACKNOWLEDGEMENT

REPUBLIC OF THE PHILIPPINES)

BEFORE ME, a Notary Public for and in the LEGAZPI CITY, personally appeared in this _____ day of _____ 2018 at the following:

| NAME | ID NO. | ISSUED BY |
|---|---|---|
| ANTONIO A. ABAWAG | QIEC-DENRB-PIR-3-4-1998 | DENR-Dil., QC |
| RAUL G. BRADECINA | EC6519477 | DFA LEGAZPI, JAN. 19, 2016 |
| DENNIS V. DEL SOCORRO | RS-001 | BFAR C.O., Q.C. |

Known to and to me the same persons who executed the foregoing instrument and they acknowledge to me that same is their free and voluntary act and deed as wee as that entities they respectfully represent.

WITNESS MY HAND AND SEAL on the date and place first above-written.

Doc. No. 343
Page No. 71
Book No. 80
Series of 2018

ATTY. ERLINDA ACHACON-MURO
NOTARY PUBLIC
UNTIL DECEMBER 31, 2019
ISSUED AT LEGAZPI CITY
ROLL OF ATTORNEYS NO. 59
MCLE No. VI-0005903 01-17-18

Republic of the Philippines
**PARTIDO STATE UNIVERSITY**
**Camarines Sur**

## *Office of the University President*

October 20, 2017

**CRISANTA MARLENE P. RODRIGUEZ**
Regional Director
Department of Environment and Natural Resources Region V
Legaspi City

Dear **Director Rodriguez:**

In view of the PSU-DENR-BFAR collaborative research project on "*Mapping of Coastal Habitats for Potential Ecotourism Livelihood of Malabungot Protected Landscape and Seascape*" at Garchitorena, Camarines Sur, we would like to request for downloading of fund for the reimbursement of project expenditure in the amount of **Seven Hundred Thousand Four Hundred Pesos Only (PhP 710,400.00)** to be deposited in the PSU Trust Fund Regular 2082-1016-78 at Land Bank Goa.

The requested amount is consistent with the letter dated September 29, 2017 as per DENR Obligation Request and Status No. 0110110120017-09-003177 dated September 26, 2017 and Certificate of Availability of Funds in the amount of PhP1,300,000.00 for the above research project.

Very truly yours,

**RAUL G. BRADECINA, Ph. D.**
SUC President III

Received:
MA. ROSA VIDA
C. ONTOLAN

| I. Professional Services | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Name | Roles | Rate/month | Man Months | DENR Fund | PSU-CWC | | | | |
| PSU (R.A. Dioneda Jr.) | Program Leader/Proj 3 leader | 11600 | 5 | 58,000.00 | | 5 | 58000 | | |
| PSU (Patricia Candelaria) | Program Coordinator/proj 4 leader | 8800 | 5 | 44,000.00 | | 5 | 44000 | | |
| PSU (Buetre) | Project 2 leader | 8800 | 5 | 44,000.00 | | 5 | 44000 | | |
| PSU (X.J.A.Dioneda) | Proj 1 leader | 8800 | 5 | 44,000.00 | | 5 | 44000 | | |
| PSU (Irene Daet) | Project staff L1 | 4800 | 5 | 24,000.00 | | 5 | 24000 | | |
| PSU (Menchie Del Castillo) | Project staff L1 | 4800 | 5 | 24,000.00 | | 5 | 24000 | | |
| PSU (Luisa Lanciso) | Project Staff L2 | 6000 | 5 | 30,000.00 | | 5 | 30000 | | |
| PSU (Arnel Zarcedo) | Project 6 leader | 8800 | 2 | 17,600.00 | | 2 | 17600 | | |
| PSU (Leih Ann Odono) | Proj Staff level 1 | 4800 | 3 | 14,400.00 | | 3 | 14400 | | |
| PSU (Jude Villar) | Proj Staff level 1 | 4800 | 3 | 14,400.00 | | 3 | 14400 | | |
| PSU (Joseria Penino) | Proj Staff level 1 | 4800 | 3 | 14,400.00 | | 3 | 14400 | | |
| PSU (Charlie Balagtas) | Project staff L2 | 6000 | 5 | 30,000.00 | | 5 | 30000 | | |
| PSU (Jasper Nieves) | Project staff L1 | 4800 | 3 | 14,400.00 | | 3 | 14400 | | |
| PSU (J. Balasta) | Project staff L1 | 4800 | 3 | 14,400.00 | | 3 | 14400 | | |
| PSU (Marjorie Atole) | Project Staff L1 | 4800 | 2 | 9,600.00 | | 2 | 9600 | | |
| PSU (Arnold Sabuco) | Admin Staff L1 | 4800 | 2 | 9,600.00 | | 2 | 9600 | | |
| PSU (Socorro Cam) | Admin Staff L1 | 4800 | 2 | 9,600.00 | | 2 | 9600 | | |
| PSU (Perlita Manago) | Admin Staff L1 | 4800 | 2 | 9,600.00 | | 2 | 9600 | | |
| PSU (Jefferson Coruno) | Project staff L2 | 1500 | 2 | 3,000.00 | | | | | |
| BFAR (Noel Ombao, corals) | Support staff | 1000 | 2 | 2,000.00 | | | | | |
| Danlops Lopez | Support staff | 1000 | 2 | 2,000.00 | | | | | |
| Garner Bordeos | Support staff | 750 | 2 | 1,500.00 | | | | | |
| Christian Rinon | Support staff | 750 | 2 | 1,500.00 | | | | | |
| Maydelyn Vinas | Project staff L1 | 1000 | 2 | 2,000.00 | | | | | |
| BFAR (Miles, SGSW) | Technical Specialist | 15000 | 2 | 30,000.00 | | | | | |
| A. M. Mendoza Jr. | Technical specialist | 15000 | 3 | 45,000.00 | | | | | |
| V. S. Soliman | Technical specialist (coastal ecology) | 15000 | 4 | 60,000.00 | | 4 | 60000 | | |
| R.R. Dioneda Sr. | Technical spectalist (socio-economics) | 15000 | 4 | 60,000.00 | | 4 | 60000 | | |
| R.Bradicena | | | Sub-total | 647,400.00 | | | 560,400.00 | | |
| II. Maintenance and Other Operating Expenses (MOOE) | | | | | | | | | |
| *Supplies* | | | | | | | | | |
| Office supplies | | | | | | | | | |
| Field sampling paraphernalia | Transect lines, slates, pencils, stainless steel, cement and sand, alum. plates, compass, gps, underwater camera, fins, solar power set-up (bat, charger, panel) | | | 10,000.00 | | | | | |
| | | | | 50,652.00 | | | | | |
| *Traveling expenses* | 20 personsxP800/dyxs dyxx3 months | | | 222,200.00 | | | | | |
| Representation Allow. | Boat hires and other incidental expenses | | | 30,000.00 | | | 100,000.00 | | |
| *Job Order/Contract of Service/Other services* | | | | 20,000.00 | | | 20,000.00 | | |
| Cateringservices | Training, Start-up, Mid and terminal presentation workshops | | | 65,000.00 | | | | | |
| Laboratory anaylsis | water samples, fabrications, etc... | | | 15,000.00 | | | 30,000.00 | | |
| Accomodation | For guests during presentation workshop | | | 69,800.00 | | | | | |
| Survey | Administration of questionniares | | | 120,000.00 | | | | | |
| PSU-CWC | Sci. Res. Assistant | 19814.4 | | - | | | | | |
| | Science Aide | 13989.6 | 5 | 69,948.00 | | | | | |
| Admin Cost | | | Sub-Total | 652,600.00 | | | | | |
| | | | TOTAL | 1,300,000.00 | 200,000.00 | | 150,000.00 | 710,400.00 | |
| | | | | | | | | 54.64615385 | |

boat hire

Prepared by:

RONNEL A. DIONEDA JR.
Program Leader

DIR. DENNIS DEL V. SOCORRO
Regional Director, BFAR ROV

RAUL E. BRADICINA, Ph.D
SUC President III, PSU

DIR. CRISANTA MARLENE P. RODRIGUEZ
Regional Director, DENR ROV

/14 567353 2373.73 68 DEBIT ON-US 1247889.05
PF1-Fwd PF2-Bkwd PF12-Help PA2-Prompt 1245515.32

mmand ===> IMII

Page 2 12/06/17
10:52:30

DDA STATEMENT HISTORY

RTIDO STATE UNIVERSITY-TRUST FUND Account [redacted]
N JUAN BAUTISTA GOA Date Last Statement 11/30/17
MARINES SUR 4422 Date This Statement 12/06/17

*** DDA TRANSACTIONS ***

| Beginning Balance | Checks/Other Debits Nbr | Total Amount | Deposits/Other Credits Nbr | Total Amount | Ending Balance |
|---|---|---|---|---|---|
| 1235791.66 | 10 | 32985.15 | 11 | 776662.50 | 1979469.01 |

| e | Check Nbr | Amount | TP | Transaction Description | Balance |
|---|---|---|---|---|---|
| 14 | 567353 | 2373.73 | 68 | DEBIT ON-US | 1245515.32 |
| 14 | 567351 | 1402.80 | 83 | LOCAL ICC | 1244112.52 |
| 15 | | 8000.00 | 11 | CASH DEPOSIT | 1252112.52 |
| 15 | 567335 | 2273.75 | 83 | LOCAL ICC | 1249838.77 |
| 16 | | 6000.00 | 11 | CASH DEPOSIT | 1255838.77 |
| 22 | | 14000.00 | 11 | CASH DEPOSIT | 1269838.77 |
| 26 | | [redacted] | 50 | CM PYRL/OTHERS | 1970238.77 |
| 27 | | 9000.00 | 11 | CASH DEPOSIT | 1979238.77 |
| 28 | | 262.50 | 12 | ON-US CHK DEPOSIT | 1979501.27 |
| 28 | | 3000.00 | 11 | CASH DEPOSIT | 1982501.27 |
| 28 | 567361 | 3032.26 | 83 | LOCAL ICC | 1979469.01 |

?1-Fwd PF2-Bkwd PF12-Help PA2-Prompt

mand ===> IMII

Page 3 12/06/17
10:52:31

DDA STATEMENT HISTORY

TIDO STATE UNIVERSITY-TRUST FUND Account 000-208-0000-20821016?8
JUAN BAUTISTA GOA Date Last Statement 11/30/17
ARINES SUR 4422 Date This Statement 12/06/17

*** DDA TRANSACTIONS ***

| Beginning Balance | Checks/Other Debits Nbr | Total Amount | Deposits/Other Credits Nbr | Total Amount | Ending Balance |
|---|---|---|---|---|---|
| 1235791.66 | 13 | 67182.42 | 11 | 776662.50 | 1945271.74 |

| | Check Nbr | Amount | TP | Transaction Description | Balance |
|---|---|---|---|---|---|
| 8 | 567361 | 3032.26 | 83 | LOCAL ICC | 1979469.01 |
| 8 | 567336 | 32273.22 | 83 | LOCAL ICC | 1947195.79 |
| 9 | 567359 | 857.25 | 83 | LOCAL ICC | 1946338.54 |
| 9 | 567360 | 1066.80 | 83 | LOCAL ICC | 1945271.74 |

Republic of the Philippines
DEPARTMENT OF ENVIRONMENT AND NATURAL RESOURCES
Regional Office No. V, Rawis, Legazpi City
☎ (052)482-0697 /482-0691 ☎ 482-0693

Project Title : **COASTAL AND MARINE ECOSYSTEM MANAGEMENT PROGRAM (CMEMP) FOR MALABUNGOT PROTECTED LANDSCAPE AND SEASCAPE.**

Location : Garchitorena, Camarines Sur

Area : 120.62 hectares

Project Cost : Php 1,300,000.00

## TERMS OF REFERENCE

**A.** The **Coastal and Marine Ecosystem Management Program (CMEMP)** in the Malabungot Protected Landscape and Seascape (MPLS) shall be implemented with the activity entitled "Applied *Research on the Mapping of Coastal Habitats for Potential Ecotourism Livelihood*". This activity shall be done thru partnership with other agencies and the academe as convergence initiatives. It shall also seek the assistance of consulting services as prescribed under the Government Procurement Reform Act otherwise knnown as R.A. 9184.

**B. Project Duration**: until December 2017

**C. Scope of Work**

C.1. Socio Economic Activities

a. Training Need Assessment (TNA)
b. Socio Demographic
c. Tenure
d. Perception and attitude towards CRM, Ecotourism and Ressetlement
e. Existing livelihood

C.2. Ecological Assessment

a. Mapping of Coastal Habitats – survey, GIS mapping, and ground truthing. Survey shall be conducted using a precise survey instrument to come-up with digital thematic maps using updated geographic information system (GIS) software. Among Others, the following maps are required: individual maps of the mangrove forest, seagrass beds, coral reefs, map of the Malabungot Protected Landscape and Seascape (MPLS)

b. Habitat Assessment – coral reef and reef fishes (FVS), Seagrass and seaweeds and macroinvertebrates, and Mangroves and cryptobiota Capture Fisheries. Habitat assessment for coral reefs, mangrove forests

and seagrass beds shall be conducted to determine extent and status of these ecosystems. Methodology for habitat assessment should be consistent with the guidelines of the Biodiversity Management Bureau (BMB) Technical Bulletin No. 5 dated February 08, 2017.

An Autonomous Reef Monitoring Structure (ARMS) will be fabricated to determine the biodiversity of the coral reefs. The ARMS will be deployed within the marine protected area and will be analyzed and monitored regularly.

C.3. Total Economic Valuation

C.4. Ecotourism Development Planning Project

a. Toursim Structure and Tourism Product Development Plan
Key Acitivities
a.1 PRA and FGD for tourism attraction scoping
a.2 Workshop for tourism structure unit and proposed activities
a.3 PRA and FGD for Community Task Mapping
a.4 Workshop for Community Task Development
a.5 Survey of Tourist Attitude and Preference
a.6 Tourism Product Development and Segmentation Workshop

b. Tourism Livelihood Development Plan
Key Activities
b.1 Participatory Resource Inventory/Assessment for livelihood identification
b.2 Liveihood and Rural Enterprise Development Workshop (input also survey result on existing livelihood)

c. Tourism Skills and Capability Development Plan
Key Activities
c.1 Workshop for capacity training and development plan

d. Tourism Sustainble Financing and Resource Governance Plan
Key Activities
d.1 Valuation of Ecosystems: WTP survey
d.2 Workshop for Establishing PES and TEV
d.3 PRA for Community CRM
d.4 CRM Planning by communities workshop
d.5 Workshop for Integration of Community CRM plans to LGUs/Sensegu level

e. Tourism Carrying Capacity and Monitoring System and Structure Developmen
Key Activities
e.1 Carrying capacity survey
e.2 Workshop for developing carrying capacity
e.3 Workshop for identifying parameter and status for biodiversity andhabitat monitoring system and structure

Ecotourism livelihood planning should be sonsistent with DAO 2013-19 or the Guidelines on Ecotourism Planning and Management in Protected Areas.

A coastal resources and marine ecosystem management plan should also be formulated for *Barangay Binagasbasan*, Garchitorena, Camarines Sur.

**D. Execution of Work**

a. Upon approval of the Memorandum of Agreement, the winning PO/NGO shall immediately commence the activities indicated in the approved Work and Financial Plan in coordination with the Office of the DENR for proper supervision and monitoring of the project.

b. Regular monitoring of the project activities shall be conducted by the technical personnel of the DENR from 1. CENRO/PENRO Level 2. Regional Office Level and 3. National Level to ensure that all activities are performed in accordance with the approved Work and Financial Plan.

c. The Contractor shall maintain records of all activities being performed and be made available at all times to the DENR staff.

d. DENR Composite Inspection and Monitoring Team shall conduct inspection of the accomplished activities upon receipt of the submitted accomplishment report from the winning bidder and recommend payment corresponding to the activities actually accomplished.

**E. Payment Schedule**

A total budget of Php1,300,000.00 will be alloted for this activity. Payment schedule will be through actual activities accomplished or progress billing on a reimbursement scheme, consistent with the rules and regulations setforth under R.A. 9184 or the Government Procurement Reform Act as validated by a Composite Inspection and Monitoring Team from the DENR.

Prepared by:

**MA. ROSA VIDA C. ONRUBIA**
Chief, Coastal Resource & Foreshore Management Section

Reviewed by:

**GRACE L. CARINO**
Chief, Conservation & Development Division

Recommended by:

**ROBERTO D. SHEEN**
Asst. Regional Director for Technical Services

Approved by:

**CRISANTA MARLENE P.RODRRIGUEZ**
Regional Director

PARTIDO STATE UNIVERSITY
Goa, Camarines sur

"Applied Research on Mapping of Coastal Habitats for Potential Ecotourism livelihood in Malabungot Protected Landscape and Seascape", Binagasbasan, Gachitorena, Camarines Sur

**Financial Status**
as per Program of Receipts and Expenditure
FY 2017-2018
First Tranche

| OBJECT OF EXPENDITURE | BUDGET | DISBURSEMENT | OUTSTANDING BALANCE |
|---|---|---|---|
| **Personal Services** | | | |
| Honoraria - | 369,200.00 | 189,600.00 | 179,600.00 |
| **Total Personal Services** | 369,200.00 | 189,600.00 | 179,600.00 |
| | | | - |
| **Maintenance and Other Operating Expenses** | | | - |
| Supplies | - | | - |
| Office Supplies & Field Paraphernalia | - | | - |
| Traveling Expenses | 180,700.00 | 76,796.00 | 103,904.00 |
| Job Order/COS/Other Services | - | | - |
| Catering Services | - | | - |
| Laboratory Analysis | - | | - |
| Accomodation | - | | - |
| Survey | 107,100.00 | | 107,100.00 |
| Science Research Assistant | 43,400.00 | | 43,400.00 |
| **Total MOOE** | 331,200.00 | 76,796.00 | 254,404.00 |
| **Total PS and MOOE** | 700,400.00 | 266,396.00 | 434,004.00 |

1033,72.14
369,796.14 = 3300

Prepared by:

**CRISTINE P. PEJO**
Accountant In-Charge
Trust Fund

Partido State University
Goa, Camarines Sur

"Applied Research on Mapping of Coastal Habitats for Potential Ecotourism livelihood in Malabungot Protected Landscape and Seascape",Binagasbasan, Garchitorena, Camarines Sur

Expenses as of March 31, 2018

| **FUND TRANSFER** | | | | | | 700,400.00 |
|---|---|---|---|---|---|---|
| **Disbursement** | | | | | | |
| **Date** | **Check No.** | **DV No.** | **Payee** | **Particulars** | **Amount** | |
| 12/28/2017 | 567377 | 120121 | Patricia M. Cadelari | Reimbursement for Boat Rental for the | 7,020.00 | |
| 12/28/2017 | 567378 | 120122 | Irene Daet | Reimbursement for Traveling Expenses | 3,372.00 | |
| 12/28/2017 | 567379 | 120123 | Menchi E. Del Castill | Reimbursement for Traveling Expenses | 2,500.00 | |
| 12/28/2017 | 567380 | 120125 | Josenia Penino | Reimbursement for Traveling Expenses | 2,500.00 | |
| 12/28/2017 | 567381 | 120124 | Redentor Buetre | Traveling Expenses-Mbalungot Project | 3,086.00 | |
| 12/28/2017 | 567382 | 120126 | Patricia M. Cadelari | Reimbursement for Traveling Expenses | 4,340.00 | |
| 12/28/2017 | 567384 | 120128 | Raul G. Bradecina | Reimbursement for Traveling Expenses | 2,560.00 | |
| 12/28/2017 | 567385 | 120129 | Joel Balasta | Traveling Expenses-Mbalungot Project | 7,202.00 | |
| 12/28/2017 | 56786 | 120130 | Leih Anne Odoño | Reimbursement for Traveling Expenses | 2,560.00 | |
| 12/28/2017 | 56787 | 120131 | Ronnel A. Dioneda J | Reimbursement for Traveling Expenses | 2,560.00 | |
| 12/28/2017 | 56788 | 120132 | Luisa M. Lanciso | Reimbursement for Traveling Expenses | 2,560.00 | |
| 12/28/2017 | 56789 | 120133 | Xavier Jules A. dione | Reimbursement for Traveling Expenses | 3,016.00 | |
| /16/2018 | 567427 | 10021 | Redentor Buetre | Honararia-Malabungot Project | 8,800.00 | |
| /16/2018 | 567428 | 10022 | Jasper Nieves | Honararia-Malabungot Project | 12,000.00 | |
| 1/16/2018 | 567429 | 10023 | Irene P. Daet | Honararia-Malabungot Project | 4,800.00 | |
| 1/16/2018 | 567430 | 10024 | Leih Ann R. Odoño | Honararia-Malabungot Project | 4,800.00 | |
| 1/16/2018 | 567431 | 10025 | Joel Balasta | Honararia-Malabungot Project | 4,800.00 | |
| 1/16/2018 | 567432 | 10026 | Chalie V. Balagtas | Honararia-Malabungot Project | 4,800.00 | |
| 1/16/2018 | 567433 | 10027 | Patricia M. Candela | Honararia-Malabungot Project | 17,600.00 | |
| 1/16/2018 | 567434 | 10028 | Ronnel A. Dioneda J | Honararia-Malabungot Project | 23,200.00 | |
| 1/16/2018 | 567435 | 10029 | Luisa M. Lanciso | Honararia-Malabungot Project | 12,000.00 | |
| 1/16/2018 | 567436 | 10030 | Menchi Del Castillo | Honararia-Malabungot Project | 4,800.00 | |
| 1/16/2018 | 567437 | 10031 | Josenia Penino | Honararia-Malabungot Project | 4,800.00 | |
| 1/16/2018 | 567438 | 10032 | Raul G. Bradecina | Honararia-Malabungot Project | 30,000.00 | |
| 1/16/2018 | 567439 | 10033 | Socorro C. Cam | Honararia-Malabungot Project | 4,800.00 | |
| 1/16/2018 | 567440 | 10034 | Perlita D. Mañago | Honararia-Malabungot Project | 4,800.00 | |
| 1/16/2018 | 567441 | 10035 | Xavier Jules A. Dione | Honararia-Malabungot Project | 17,600.00 | |
| 1/18/2018 | 567448 | 10050 | Ronnel R. Dioneda S | Honararia-Malabungot Project | 30,000.00 | |
| 12/28/2017 | 567383 | 120127 | Charlie Balagtas | Reimbursement for traveling expe | 2,560.00 | |
| 3/2/2018 | 567466 | 30075 | Joel Balasta | Cash Advance-Traveling Expenses | 3,440.00 | |
| /3/2018 | 567467 | 30076 | Jasper Nieves | Cash Advance-Traveling Expenses | 3,440.00 | |
| /4/2018 | 567468 | 30077 | Marjorie C. Afole | Cash Advance-Traveling Expenses | 3,440.00 | |
| 3/5/2018 | 567469 | 30078 | Percival Ebron | Cash Advance-Traveling Expenses | 3,440.00 | |
| 3/6/2018 | 567470 | 30079 | Menchi E. Del Castill | Cash Advance-Traveling Expenses | 3,440.00 | |
| 3/7/2018 | 567471 | 30080 | Ronnel A. Dioneda J | Cash Advance-Traveling Expenses | 3,440.00 | |
| 3/8/2018 | 567472 | 30081 | Arnold Sabuco | Cash Advance-Traveling Expenses | 3,440.00 | |
| 3/9/2018 | 567473 | 30082 | Xavier Jules A. dione | Cash Advance-Traveling Expenses | 3,440.00 | |
| 3/10/2018 | 567474 | 30083 | Patricia M. Candela | Cash Advance-Traveling Expenses | 3,440.00 | |
| **Total** | | | | | **266,396.00** | |
| **FUND BALANCE** | | | | | | **434,004.00** |

| March 2018 Honoraria | |
|---|---|
| Nieves | 6,000.00 |
| Candelaria | 8,800.00 |
| Ronnel Jr. | 11,600.00 |
| Balasta | 4,800.00 |
| Jules | 8,800.00 |
| Travelling Expenses | |
| April 28-May 1 Travelling Expenses | 26,400.00 |
| April 28-May 1 Boat Rental | 14,000.00 |
| March - April R.A | 22,972.14 |
| TOTAL | 103,372.14 |

1

~~8~~ 22

1706

1700

1800

5

20

21

## I. Basic Information

| | |
|---|---|
| Research Program Title: | Assessment, Total Economic Valuation of Coastal Habitats and Development of Sustainable Ecotourism Master Plan for Malabungot Protected Landscape and Seascape |
| Implementing Agencies: | Dept. of Environment & Natural Resources ROV (DENR-V), Legazpi City<br>Partido State University-Coastal Wetland Center (PSU-CWC), Sangay, Camarines Sur<br>Bureau of Fisheries and Aquatic Resources V (BFAR V), Bula, Camarines Sur<br>Bicol University-Center for Policy Studies and Development (BUCPSD, Legazpi City<br>Local Government of Garchitorena, Province of Camarines Sur |
| Key Proponent: | Ronnel A. Dioneda Jr.<br>Faculty-Researcher, PSU-CWC, PSU Sangay Campus<br>CP# 09272470226 email add: rondioneda@gmail.com |
| Duration: | August 2016 to January 2017 |
| Funding Requirement: | **PhP 1,500,000.00**<br>P1.3M from DENR-V and P200K from PSU-CWC |
| Funding Source | DENR RO V under its Coastal and Marine Ecosystem Management Program (CMEMP)<br>Partido State University-CWC |

## II. Rationale

The protected area system has long been practiced in the country primarily as part of the multitude of interventions to preserve natural endowments and to put in check degradations due to natural and anthropogenic activities. The Philippine Commission Act No. 648 (passed by the US congress in 1903) marked the pioneering foundation of the national parks and protected area initiatives in the country. The scope of these protected areas was expanded by Act No. 1148, also known as Forest Act in 1904. This later on gave birth to establishment of national parks by virtue of the enactment of the National Parks Act (Act No. 3195) in 1932. This was followed by the revised Forestry Code in 1975 and the National Integrated Protected Area System Act (RA 7586 of 1992). The later provides the legal framework for the establishment and management of protected areas in the Philippines. The NIPAS Act is part of the actualization of the commitment of the Philippine government in various international conventions and agreements it entered into. Under the NIPAS, protected areas in the country are classified and managed in varied ways. One of them are natural parks. NIPAS describe protected landscape/seascapes as areas of national significance which is characterized by the harmonious interaction of man and land (and the seas) while providing opportunities for public enjoyment through recreation and tourism within the normal lifestyle and economic activity of the area. Pursuant to NIPAS, the Malabungot Protected Landscape and Seascape (MPLS) was established in on April 23, 2000 under Proclamation No. 288, which was later on formally legislated under the senate bill no. 2895. The protected area is located at Malabungot Island of the municipality of Garchitorena and encompasses an area of and formally 120.62 hectares. There is no available baseline records that were used in establishing the area as NIPAS protected area nor any recent technical information showing monitoring. Around eight kilometers Northeast of Chico Island is a network of smaller MPAs established by Garchitorena LGU two years ago.

The sustainable utilization of critical ecosystems and habitats which provide the *reason d' etre* of seascape and landscape conservation enabling acts is commonly challenged by keeping a balance with its resource and social components. To achieve sustainability the management of these triad components as systems becomes imperative. Deeper understanding of the status of these systems and their dynamic interactions are key requirements for their rational management in the context of protected seascapes and landscapes.

For instance, corals, seagrass and seaweeds and mangroves are considered life support systems in the coastal zones. These aquatic habitats are production systems that produce fishery resources. However, anthropogenic activities and natural disturbances create a synergy of impacts that cast a cloud of uncertainty to their long-term existence and the provision of benefits from these habitats. Alcala (1998) demonstrated that a healthy reef system could provide as much as 18 mt/km$^2$/year while those in poor state could measly just provide 2 mt/km2/year. Considering the scarcity of technical baseline information for MPLS, there is a dire need to comprehensively assess its coastal ecology. The management of the MPLS and the attached resources should be well-supported with scientific information.

Coastal ecosystems and habitats, aside from the resources they produce, also provide tremendous benefits in forms of ecological services. The very strong reliance of people from marine ecosystems' benefits erodes both the quality and quantity of the services provided by these ecosystems. This is in the face of fast degradation of many coastal zones by the combined impacts of natural and intense man-made disturbances that are further compounded by climate change impacts. Commonly, these ecological services don't have monetary values as they don't have markets. The absence of monetary values imputed by society for these services results to inadequate investments of public funds by decision makers to promote their ecological integrity and health. Scholars recognize the fact that the degree at which these ecosystems effectively function and provide goods and services are dependent on their integrity and health (Calumpong et al., 2013). This underscores the need for the economic valuation of the services given by these ecosystems to generate the compelling basis for investing public funds for their sustainable management. Recent local studies found out that the economic value of goods and services from coastal resources in the five SCREMP sites in Albay is tremendous. Torres et al (2015) estimated annual foregone benefits could run to PhP 238 millions of pesos if the present poor too fair conditions of its coastal habitats prevail.

Over years, it has been observed that communities in fishing villages are the poorest. Social scientists attribute this observation to the open access nature of the resource as well as the lack of opportunities brought by the geographical remoteness of most fishing villages to economic hubs. The open access nature of the coastal resources results to the occurrence of the seminal tragedy of the commons in which common resource owned by everybody brings ruin to all. Also, the lack of opportunities in coastal fishing villages results to resource dependency, unsustainable resource use and economic vulnerability. While the establishment of institutions such as MPAs and enabling acts that declare coastal and marine areas as protected seascapes and landscapes served as management tools to address the tragedy of the commons, the challenge for developing sustainable financing mechanisms to incentivize resource users and stakeholders to support and promote sustainable resource governance weakens their effectiveness. This provides the imperative for establishing a payment for environmental services system in the form of users' fees. Whereas financial assistance and livelihood supports have been promoted to fishing communities in the past to uplift their living conditions, their being resource-based and extractive nature compounded by market accessibility constraints had casted cloud of doubts on their viability and sustainability.

Recently, sustainable ecotourism has been increasingly recognized to have the potential to address the problem brought by lack of opportunity in many fishing communities brought by its geographical remoteness. Living up to the mantra of "leaving nothing, except footprints; taking nothing, except memories", this innovative form of social enterprise aptly harmonizes with remote fishing communities as non-extractive livelihood strategy because of its ability to use tourism commons as entrepreneurial commodities where the market comes to the product. A marriage between coastal resource management and sustainable ecotourism will address the twin problems

of open access nature and lack of opportunities in fishing communities that fuels poverty and resource and ecological degradation. Developing a framework and a roadmap that comprehensively integrates coastal resource and biodiversity conservation and sustainable ecotourism that are both scientific information and market-driven will not only make MPLS as a tool for protecting biodiversity and rationally managing coastal resources, but also as an engine of growth that will fuel the rural economies of the communities surrounding it.

This research program would be a pioneering intervention in the MPLS to achieve the envisioned directions. In general, it would look into the present condition of the bio-ecological, resources and socio-economic dimensions attached to the resource utilization and biodiversity conservation as valuable inputs for the rational management and establishing a roadmap for sustainable ecotourism in the MPLS and adjoining area. Specifically, the research program intends to accomplish the following:

1) Investigate the diversity, community structure, density, percent cover, frequency cover, biomass of seagrass and seaweeds of MPLS together with accounts of observable wastage;
2) Assess the health of the coral reef systems in terms of percentage cover of major lifeforms and provide accounts of observable disturbances
3) Determine the diversity, species density, basal area and other measures of community structure for mangroves and provide accounts of observable perturbations.
4) Characterize the water quality and the 24-hour plankton dynamics of MPLS;
5) Deploy autonomous reef monitoring structure at MPLS and LGU MPA;
6) Characterize the capture fisheries and the reef associated reef fishes of Garchitorena with emphasis at MPLS;
7) Generate basic bathymetric map and coastal habitat maps of MPLS with estimate of areas per habitat;
8) Generate capacity development information needs of personnel in the existing industries in the MPLS area
9) Produce the socio-demographic information, perceptions and attitude towards coastal resource management and ecotourism development, tenure, resettlement and livelihood
10) Identify the resources and habitats available in the area, the community interventions to the resources and habitats, the problems/threats related with the resources and habitats
11) Generate the monetary value imputed by society for the non-use value of the MPLS seascape
12) Provide inventory of various tourism attractions covering both natural and cultural assets within the MPLS area.
13) Generate recreational attitude, preference, travel cost profile, and market segment profile
14) Develop potential livelihood program that will provide potential development of sustainable ecotourism industry in the area
15) Identify the factors affecting the compliance and sustainability of the tourism assets in the area
16) Integrate all outputs of the Writeshops cum workshops for development planning to generate the draft of the MPLS TMP.

## III. Research Components, Methodologies and Expected outputs.

### Project 1. Coastal Ecological Assessment of MPLS (Marjorie Atole and XJ Dioneda)

This project will provide the much needed updated information regarding the status of the existing coastal habitats and the coastal environmental conditions of the protected area. The former would investigate the status of coral reefs, seagrass and seaweed beds and mangroves, using conventional ecological measures. Coastal environmental profile would look into the water quality and the plankton community it supports. Cryptobiota would also be investigated in this project using the BU-DENR-made Autonomous Reef Monitoring Structure (ARMS). Locations and extent of occurrence of the coastal habitats will be initially determined from existing records and inputs from the LGU of Garchitorena.

Study 1. Seagrass and Seaweeds (SGSW) assessment of MPLS
Researchers: PSU (X.J. Dioneda, Study Leader; BFAR Miles, and Christian)

Basic initial profiling and mapping of the location and distribution of seagrass and seaweed communities in each municipality shall be carried through reconnaissance surveys. Informal consultation with local folks shall be undertaken to ascertain and finalize the location of sampling sites. Number and location of stations shall be determined after the preliminary field survey.

Comprehensive S G S W assessment shall be undertaken using the conventional line quadrat method (LQM, English et al 1997). Minimum of three sampling stations will be assessed within the natural park. Each station shall have three 50-meters transects as replicates. Laid perpendicular to the shoreline, these three replicate transects will be laid continuously seaward whenever possible (i.e., extent of seagrass community presence seaward is 150 meters or longer). A 0.5x0.5 meter quadrats will be laid every 10 meter distance. Seagrass and seaweeds that are covered by the quadrats shall be assessed based on species presence, frequency cover and percent cover by species. If allowed for scientific research purposes, SGSW would be harvested for comprehensive assessment of species biomass estimation. The following shall be established in this study

a. Seagrass and seaweeds species observed in the sampling stations
b. SGSW species density, frequency, percent cover, canopy height for Thalasia sp.
c. Biomass per species (when sample harvesting is allowed)
d. Community structure of the SGSW measured in conventional ecological indices.
d. Accounts of the associated flora and fauna, disturbances and existing resource uses.
e. Establishment of at least six permanent monitoring stations which are geo-referenced and marked with permanent markers for long-term SG/SW monitoring.

Study 2. Coral reef Assessment of MPLS
Researchers: PSU: (R.A. Dioneda Jr. Study Leader; Noel Ombao, Member, Joel Balasta, and Garner)

Coral reefs have been traditionally protected through the marine protected area system of many local government units (LGUs). Many MPAs in the Philippines were established based on the presence and the intended protection of this coastal habitat. This can be due to the documented economic and ecological benefits they provide. In this study, the coral reef systems of Malabungot MPLS would be subjected to comprehensive assessment.

Location of the coral reef areas shall be initially determined from existing records and coordinative discussions from concerned people in Garchitorena. Validation shall be done through reconnaissance survey (manta tow technique) coupled with the traditional knowledge of local fishers. This is to determine the relative location and concentration of the coral reef systems in each municipality and help identify sampling stations. At least three stations will be identified in Malabungot reef systems. Two methodologies will be carried-out in coral assessment. First is the conventional Line Intercept Transect (point intercept) outlined by English et al (1997). This will be reinforced by coral photo transect method developed by Roelfsema and Phinn (2009) of the university of Queensland. Both methods shall be used in all stations, aided with SCUBA diving

system and shall use 50 meters transect. LIT would be carried out by videotaping the lifeforms traversed by the transect line. Reading of point intercepts would be carried-out in the laboratory and analysis would be carried-out in excel. Photo transect would use the minimum required paraphernalia (GPS with PC interface capability, wide angled lensed underwater camera, plumb line, transect line, floats, dry bag and reel). Photos will be analyzed using standardized software Coral Point Count excel (Kohler and Gill, 2006) or other simpler alternative analytical softwares.

The same set of assessment protocols would be employed in assessing the status of the coral reef systems of the three MPAs established by the LGU in Garchitorena. Although outside of the geographical coverage of the MPSL, providing technical information about the reef systems of the LGU-established MPAs would substantiate the existence of these MPAs. This is on the light of the fact that their establishment was devoid of any comprehensive technical basis (i.e., the MPAs are put in place just on the basis of coral occurrence from manta tow surveys done three years ago).

The following are the deliverables under this study:

a. Percent cover estimates of coral substrate types and benthic organisms and their criteria ratings outlined by Gomez and Alcala (1979);
b. Documentation of possible reef impacts such as cases of bleaching, poison and blast fishing, diseases, trashes, anchorage, fishing and other perturbations;
c. Geo-referenced sampling sites. Transects from selected stations shall be designated as permanent monitoring stations marked with monitoring blocks with the prescribed dimensions;
d. Geo-tagged photos of the assessment conducted; and,
e. Comprehensive report for the coral reef assessment and fish visual census

Study 3. Assessment of Mangroves at MPSL

Researchers: PSU (X.J.Dioneda: Study Leader; Members: May Vinas and Jasper Nieves)

As seen in Google earth, mangroves of MPLS appear to just occur at the southern, eastern and northern sections of Malabungot Island. Preliminary site inspection is needed to ascertain their occurrence and to be able to design appropriate sampling design. Line Plot method (English et al 1997) shall be employed to assess the mangrove communities of MPLS. At least three transects (100 meters when possible) shall be laid per mangrove cluster (i.e., southern, eastern and northern). Coordinates of stations shall be obtained and appropriate marking will be obtained for establishment of permanent monitoring stations. The following shall be obtained in this study:

a. Species and count of trees, saplings and seedlings per species (regenerative capacity)
b. Percent crown cover
c. Basal area per species and basic ecological indices of the mangrove communities per section/cluster.
d. Documentation of observable perturbation
e. Geo-tagged photos of the assessment conducted; and,
f. Comprehensive report for the mangrove assessment.

Study 4. Water quality and the plankton community of MPLS waters

Researchers :(RA Dioneda Sr. Study Leader; R.A. Dioneda, Jasper Nieves and Sci. Res. Aide)

Plankton community dynamics is strongly influenced by the day and night cycle. Species composition and abundance change at certain location and depth as plankton tend to be strongly respond to and affected by light and dark cycle and by current driven by tide and other forces. A 24-hour sampling of plankton on key station would then help describe in full detail the bio-ecological dynamics of plankton of the natural park. This study would then intend to establish the following: 1) Plankton diversity and abundance in the natural park 2) Temporal plankton profile within a 24-hour cycle 3) Document vertical migration of zooplankton in a specific station. These objectives will be achieved through a 24-hour plankton sampling. Water sample of known volume at certain depth will be collected using a Kemmerer water sampler. Samples will be taken from the bottom and with a one meter interval thereafter. Samples will be immediately filtered at 60 um (for

zooplankton) and at 20um for phytoplankton. Concentrated samples will be preserved using acid lugols solution and microscopy will be held at the PSU-CWC laboratory.

Together with the 24-hour plankton sampling is the determination of the basic physical and chemical properties of the water of a certain station in the natural park. Portable multiparameter meters will be used here to measure pH, DO, TDS, salinity, temperature and other parameters. Data loggers (Hobo) to record the temperature and light intensity will be deployed for at least two days at the bottom and near-surface locations. Light penetration will be measured using secchi disk. Some water samples will be collected and fixed for nutrient analysis (nitrate, phosphate). Water quality measures shall be compared to the DAO 2016-08 standards.

Study 5. Deployment of Cryptobiota Settlement Modules at MPLS.
Researchers: (RA Dioneda Jr, Study Leader; With assistance of other divers)

Reefs are dynamic systems which continually undergo active interplay between destructive bioerosion and constructive formation (Schuhmacher 1978). The speed at which a reef is formed depends on rates of skeletongenesis and consolidation, while destruction rates depend upon physical, chemical and biological eroding processes (Risk and Sammarco 1982, Hutchings 1986). The role of cryptic organisms on both processes have been documented to be crucial. They increase the complexity of substrates which cons-antly create spaces for coral larvae and other organisms, thereby maintaining diversity (Hutchings 1986). Taken with the degrading impacts of factors driven by climate change, fate of coral reefs maybe driven more on a destructive end. This research adopts the definition cryptic fauna (or cryptofauna or cryptobiota) of Hutchings (1978), Peyrot-Clausade (1981) and Scott (1987) which means organisms living in small pores, holes or cavities within the rock or inside organisms that cover them.

The purpose of this project is to describe the diversity ad abundance of cryptic fauna associated with coral reefs. This will also provide opportunities to test the locally designed cryptobiota settlement trap (CST). As cryptic organisms, they are less studied and are rarely investigated in conventional ecological studies. But considering the above-stated ecological importance, investigation of cryptobiota would be necessary especially to protected areas that are expected to repopulate unprotected zones by spillover or by other means.

Following the autonomous reef monitoring structure (ARMS) schematics of the Smithsonian Institute, Bicol University and DENR-V fabricated ARMS modules made of acrylic plastic. Two ARMS modules will be deployed in the reef systems of MPS and in the LGU MAP. These ARMS would be coupled with temperature data logger and would be maintained in geo-referenced locations for a year. SCUBA-aided photo documentation would be made at least two times within the project duration. Likewise, data from the logger would be downloaded and the logger would be redeployed for long-term data logging. Metagenomic studies will be employed in the future to reveal the cryptobiota assemblage of MPSL reef systems.

**Project 2. Capture Fisheries and Reef Associated Fishes of MPLS (Redentor Buitre and RA Dioneda Sr)**

Study 6. Capture fisheries of Garchitorena with emphasis on Malabungot waters.
Researchers (Redentor Buetre, Study Leader and Irene Daet, Member)

Fisheries is one of the primary extractive economic activity that impacts coastal resources' long term abundance. Its potential to decimate a fish stock is very high especially when the fishing intensity is beyond the regenerative capacity of the fish. There is then a need to characterize the fishery resource extraction in Malabungot and adjoining areas. One way to do this is to characterize the capture fisheries of the area. Along this line, catch characterization would be done by direct catch and landing observation/surveys, market surveys and recall interview with key informants. Seasonality of fishing operation, resource use conflicts/issues and other fisheries concerns would be determined by focused group discussions and through semi-structured interview. Type and number of fishing gear

implements shall be obtained by gear inventory and using the Fish and boat R data of BFAR and LGU records.

Study 7. Reef Fish Assemblage and Biomass of MPSL and Garchitorena LGU MPAs.
Researchers: PSU: (R.A. Dioneda Study Leader; BFAR: Janlops and Joel Balasta)

The variety, abundance and biomass of the coral reef fishes are directly attributed to health of reef systems. Good reef fish assemblage also support tremendous supply of fish catch reaching to about 25% of coastal fisheries productivity (Alcala and Russ 2002). There is then a need for the inclusion of the reef fish monitoring in CRM monitoring activities. Along this line, fish visual census (FVC) will be carried-out in selected stations used for coral reef assessment. Here, a diver would identify count and estimate the approximate total lengths of the reef fishes encountered along the entire stretch of the transect line used for coral assessment. This study would establish the following: 1) diversity of reef fishes in station, 2) Biomass per species and total biomass for the station, 3) notes on indicator species and 4) community structure defined by conventional ecological indices per station.

**Project 3. Coastal Habitat and MPLS Location Revalidation Mapping and Basic Bathymetry of MPSL** (RA Dioneda, Jr and Mapping expert)

Study 8. Mapping of coastal habitats of MPLS.
Researchers: DENR

Distribution and expanse of the coastal habitats shall be determined using GPS-based survey around Malabungot Island. Boat sorties equipped with GPS with PC interfacing capabilities shall be made along the edges of the reef systems, sea grass and seaweeds and the sea-ward side of mangroves. Trek by foot shall be one along the land-ward edge of the mangrove areas. These geo-referenced information shall be used for digitized mapping process using acceptable tools (GIS or manifold). This study intends to come-up with a map the coastal habitats of the MPSL with estimated of area per habitat.

Study 9. Revalidation Survey of Location and Basic Bathymetry of MPLS
Researchers: PSU (R.A. Dioneda) and DENR

The coordinates of both the Proclamation 288 and the Senate Bill No. 2895 yielded erroneous coordinates for the point 1 reference. When plotted, it's near Atulayan Island of Sangay, Camarines Sur. If the bearings and defined distances between points would be traced, it would draw a polygon inside Lagonoy Gulf, around a kilometer to the East of Atulayan Island. Being a protected area with access restrictions and regulations, problematic technical definitions of its locations would have wide ranging implications to law enforcement and management decisions. There is then a need for a revalidation study to correct this technical flaw.

To characterize the bottom topography of the immediate water of Malabungot Island, a bathymetric survey will be carried-out. A Garmin Fishfinder Striker 4 will be used to characterize bottom contour around the island. Additional GPS (Garmin 7160) units will be used to navigate preset routes. Basic depth profile (in defined isobaths if possible) around the island would be the basic output of this activity.

**Project 4. Socio-Economic Assessment (Patricia Candelaria and Luisa Lanciso)**

Study 10. Training Needs Assessment
Researchers: PSU (Patricia Candelaria, Study Leader; Menchi del Castillo, Member)

The training needs assessment survey will generate the information of the capacity development needs of personnel in the existing industries in the MPLS area. The survey will generate the following data:

Priority capacity development needs in trade skills

Priority capacity development needs in food product and development skills
Priority capacity development needs in agriculture livelihood development skills
Priority capacity development needs in fisheries livelihood skills
Priority capacity development needs in enterprise development
Priority capacity development needs in tourism management skills
Priority capacity development needs in tourism enterprise development skills
Priority capacity development needs in upland and coastal resource management skills

It will employ key informant interview as data generating tool and using key stakeholders in the industry sector as well as the LGU and NGA as key informants. The industry stakeholders key informants will include owners of existing resorts, inns, hotels, small scale, medium and large scale commercial and service-oriented establishments in the area such as parlors, barber shops, transport, repair shops, food processors and producers, tour operators, sari-sari stores and groceries.. The key informants in the LGU, and NGAs will include selected officials of the barangay, DEPED, and other NGA offices available in the area.

The information generated will be used to complement and enhance the TNA data that will be captured through household survey questionnaire. These data will serve as inputs in developing the capacity development plan of the MPLS Tourism Master Plan.

Study 11. Socioeconomic Assessment
Researchers: PSU (Irene Daet, Study Leader)

The socioeconomic assessment will generate the socio-demographic information, perceptions and attitude towards coastal resource management and ecotourism development, tenure, resettlement and livelihood. It will be implemented through a household survey using questionnaire as data generating tool.

The first part of the questionnaire will include personal information of the household respondent such as age, sex, educational attainment, number of children, years in fishing, years in farming, occupation (major, secondary and tertiary), fishing income, income from major, secondary, tertiary occupation, fishing operation expense,

The second part of the questionnaire will include household respondents' knowledge attitude, practices and perceptions. The knowledge part will include awareness on NIPAS-MPLS, attendance to trainings, awareness on the trainings conducted by NGAs, awareness of fish sanctuaries and reserves, level of awareness of MFRS, awareness of environmental laws, ordinances implemented, awareness of the effectiveness of environmental laws implemented, awareness of the current status of MPLS, factors affecting marine resources and awareness of the effect of climate change on coastal ecosystem and resources. The attitude part will include opinion on the current status of marine resources in the MPLS, measures to sustain the coastal and marine resources in the MPLS, PAMB to manage the MPLS, agency that would spearhead the protection and management activities of the MPLS. The practices shall include membership to organization, involvement in activities being done on MPLS, fishing routine, income from fishing, creation of biodiversity friendly livelihood activities, and identified livelihood activities. The perception will include perception on the quality of the coastal and marine ecosystems and habitats as well as the status of the resources 10 years from now and at present. It shall also capture their perception of the threats in terms of their scope, intensity and area covered to attribute such status. The questionnaire will adopt the insights and format of DENR knowledge attitude and practices questionnaire.

The third part of the questionnaire will include items probing on their attitude towards the coastal ecosystem and its resources as well as the directions for sustainable utilization of resources and protection of the coastal and marine ecosystems. It shall also capture their attitude towards sustainable tourism promotion and governance.

The fourth part of the questionnaire will probe on the information of their dwellings that will include, number of years residing in the area, birth place, residence of origin, status whether tenured or non-tenured migrants, size of the dwellings or lot area occupied, farm area occupied, nature of dwelling whether native, semi concrete and concrete, whether willing or unwilling to be resettled, reasons for willingness or unwillingness to resettled, perception of risk to dwelling.

The fifth part shall probe on the materials style of life of the household. This will include enumeration of the available modern day indicators of living standards such as availability of comfort rooms, power, piped water, appliances, service vehicles, communication gadgets, etc.

The sixth part will capture the current livelihood engagement of the household respondents, the available livelihood assets in the households that could be categorized under human (skills such as carpentry, baking, food processing, sculpting, etc), production assets (fishing and farming assets such as carabao, thresher, boats, nets, engine), farms, buildings, water supply, fishponds, equipment etc..

The survey uses the household as the unit of analysis. A two stage randomized probability sampling design will be adopted and the number of sample respondents will be estimated following Slovin's sample formula. In this design, the universe will be divided into rural, urban and island sites. The first stage sampling will randomly select one half of the communities in these sites. In the second stage sampling randomization, household respondents from the randomly selected communities in each site category will be randomly selected. Randomization will use the electronic record of the household head names from the data bank of COMELEC or the LGU and shall be selected through the randomization feature of Microsoft EXCEL Spreadsheet.

Study 12. Resources and Resource Use Patterns
Researchers: PSU (Luisa Lanciso, Study Leader, Member)

A PRA will generate the resources and habitats available in the area, the livelihood attached to the resources and habitats, the problems/threats related with the resources and habitats. It shall also generate seasonality of fishing, the current status of the resources and the ecosystem, the interventions done for CRM, the institutions involved, the activities of male and female members of the households.

The FGD will generate data on household, reproductive health, migration and occupation data, community projects and organization and stakeholders, environment-friendly enterprises, community and fishery organizations, influential stakeholders, community perceptions on the state of natural resources in terms of mangrove, coral reefs, seagrass, beaches, estuaries and upland forest, fishery coastal activities, number of fishers, fishing gears, boats and catch rates, coastal resource management initiatives and environmental awareness, name of marine habitats or organisms that are sensitive or vulnerable, list of coastal resources that were present before 10 years or more, fishery resources condition, most important resources (top ten fish species, crustaceans and mollusks), perceived changes in tp ten fishes, crustaceans and mullosks in the last ten years (before and after)

Two sites will be selected for the PRA and FGD, one in the island (Malabungot Island) and one in the fishing community (Bgy Binagasbasan). At least 15 key informants will be selected from representative groups of women, youth, barangay officials, elderly, NGO, academe, fishers, farmers, resort owners, private sector etc.

The PRA will employ Venn diagram, Resource Map, Gear calendar diagram, species calendar diagram, diagram for habitat and 24-hour activity, trend diagram for fish catch. The FGD will employ KI questionnaire adopting RARE PRA and FGD protocol.

**Project 5. Economic Valuation of Environmental Services and Establishment of User Fee System (Raul G Bradecina and Leih Ann Odono)**

Study 13. WTP for Protecting MPL Seascape and MPA Survey
Researchers: PSU (Lieh Anne Odono, Study Leader; Raul G Bradecina, Member)

The study will generate the monetary value imputed by society for the non-use value of the MPLS seascape. It will employ the contingent valuation method. A preliminary survey will be done to determine the minimum and maximum bids imputed by municipal resident and fishers for the protection of MPLS ecosystem. Final bids will be determined and a questionnaire survey will be developed using single bounded dichotomous choice bid elicitation approach. Respondents will be selected following randomization using Slovin formula. A two stage probability sampling will be done to select the estimated number of respondents. The first stage will identify randomly the urban

and rural community sites from the total list of the barangays in the municipality. The second stage will identify the respondents in the randomly selected sites. Data will be analyzed using NLOGIT software to determine MWTP and the parameter estimates of the WTP curve.

Study 14. Establishing User Fee System
Researchers: PSU (Atty. Jude Villar, Study Leader)

This study will determine the acceptability of establishing user fee system, identify what mechanism should be established to collect the user fee, the collecting entity and how the process by which the collected fees shall be disbursed or allocated. These data will be generated through survey and workshops with stakeholders. The question items for the data will be integrated in the WTP survey. In addition, the data will also be captured through writeshop using FGD. The participants in the FGD will be selected from representative groups of resort owners, shop owners, LGU officials, NGO, NGAs, tourists, fishers and farmers, academe.

Study 15. Total Economic Valuation of MPLS
Researchers: PSU (Raul G Bradecina, Study Leader)

This study will estimate the total monetary value of MPLS using the total economic valuation concept. This will include the estimation of the use value and the non-use value of the MPLS. The use value will focus on the resource of the coastal and marine ecosystems with markets. This will include the recreational value, as well as the economic production of the habitats captured through fishing and similar extraction economic activities. The non-use value will focus on the environmental services of the MPLS that don't have market. This will include the value of protecting MPAs and seascape for future use.

The study will use the data generated from capture fisheries surveys and FISH R data bank to analyze and estimate the use-value of the resource. It shall also impute the recreational value of the resource generated from the recreational tourism attitude and preference survey focusing on the travel costs given by respondents. The non-use value of environmental services of MPLS will be captured using CVM. This will generate the mean and aggregated WTP for protecting the MPA and seascapes.

**Project 6. Ecotourism Development Research (Charlie Balagtas and Josenia Penino)**

Study 16. Tourism Attraction Inventory
Researchers: PSU (Josenia Penino, Study Leader)

This study will generate the various tourism attractions covering both natural and cultural assets within the MPLS area. These attractions will be identified within each barangay within the MPLS area. Document analysis, key informant interviews and FGD will be held to elicit attractions with ecotourism potential and provide brief description on these tourism assets. The document analysis will involve several review activities of the Municipal Development Plan and Profile, Previous Studies and Comprehensive land Use Plans, Municipal Tourism Development Plan to determine existing and potential tourism attractions and sites. The key informant interview will also inquire from residents the areas and assets in their localities that have tourism promise. The FGD will also request the participation of municipal officials and fisher representatives to identify and locate key attractions in the MPLS that are only known to them.

Study 17. Recreational Attitude, Preference and Travel Cost Study
Researchers: PSU (Charlie Balagtas, Study Leader; Raul G Bradecina, Member)

This study will generate the tourist attitude and preferences, the frequency of visit, potential expenditures that will be incurred from the travel. It wil also generate the recreational value of the MPLS tourism and the determinants of tourists' willingness to visit MPLS. This will be generated through a survey using a questionnaire.

The first part of the questionnaire will include personal information, income and occupation and the residence of the respondents.

The second part of the questionnaire will include information on tourists travel attitude. This will cover the data on respondents willingness to travel to the place and the reasons for the decision to travel, the most likely months, weeks and days in a year he would like to travel, the frequency of travel, the expenses he would most likely spend, whether travelling alone, with companion, or family, the number of days he would like to stay, the type of transport and route preferred, the accommodation preferred and reasons for the answers.

The third part of the questionnaire will include information on tourists' preference on tourism destinations/attractions, experience and accommodation. These will include data on the preferred qualities of tourism destinations, experience and accommodation. Quality indicators for destinations, experience and accommodations will be generated and shall be assigned rank using scales of 1-5 to measure level of preference. The destinations will cover enumeration and selection options of various MPLS attractions including natural habitats such as coral reefs, seagrass, mangroves; including experiences such as boating, snorkeling, sand strolling, beach sports, game fishing, swimming, beach picnics, vacations, adventures; including cultural attractions such as community visits, barangay museums, local festivals, churches, life in the rural coastal communities, livelihood, agritour, and culinary.

The survey will involve a total of 300 respondents to be interviewed. The respondents will cover potential local tourists. Of the 300 respondents, 90 will be interviewed from tourists visiting Caramoan, Camarines Sur (to represent national tourists). The remaining 210 respondents will be distributed to respondents from selected towns of Camarines Sur namely Goa-Tigaon (70), Pili-Naga (70), and Iriga-Nabua (70). Enumerators will be hired and trained prior to the survey.

The data will be analyzed using descriptive statistics to characterize attitude and preferences of potential tourists. The recreational value of the MPLS and the factors that influence the tourist's decision to visit will be determined using zonal travel cost method following the routines of NLOGIT econometric software.

Study 18. Tourism Structure Development
Researchers: PSU (Arnel Zarcedo, Study Leader)

This study will develop in details the tourism structure of MPLS. It will identify the various tourism sites where tourism development areas will evolve by order of development priorities, the designated tourism development clusters, the themes to be developed for each cluster or tourism development area, the amenities required, and the structure of tourism circuits. This will be developed though a preliminary tourism structure development Writeshop. In this activity, the key findings in studies 17, 18 and 19 will serve as inputs. Other documents such as PDA Tourism Master Plan, Bicol Regional Development Plan, and the Garchitorena Municipal Development Plan and Profile will also be used as basis in the analysis. The tourism structure development framework prescribed by the DOT will be adopted during the Writeshop.

A workshop for finalizing the draft tourism structure will be conducted. In this activity, the draft tourism structure will be presented as working paper to the participants to facilitate discussion and development of outputs. The participants will review and revise the draft basing from their perspectives and insights as well as personal knowledge of development directions in their own areas of concerns. Participants to the workshop will be composed of tourism officers, municipal planning officers and committee chairman for tourism and infrastructures of legislative arms of LGUs, PDA, DENR, DPWH and DOT representatives, and academe.

Study 19. Tourism Carrying Capacity and Coastal & Marine Environment Quality Indicators Assessment
Researchers: PSU (Arnold Sabuco, Study Leader); Jett Atole, Member)

This study will generate the data for the carrying capacity of selected identified key tourism sites in the MPLS area as well as current status of key coastal and marine environment quality parameters such as pH, Turbidity, salinity, dissolved oxygen etc.

Carrying capacity will be generated using GIS due to time and resource constraints inherent in the project schedules. However, if time permits, "ground truthing" will be done for GIS-based parameters by field researchers. Constant values that are required in the equation for the estimation

of carrying capacity will be adopted from existing literature due to the fact that the information will be developed on an ex-ante basis that is before actual occurrence of tourism in the area.

Quality indicators will be generated through two-shot samples using multi-parameter kits to generate benchmark data. Other parameters will be used from the data generated during the ecological survey.

**Project 7. Sustainable Ecotourism Planning (Arnel Zarcedo and Raul G Bradecina)**

Workshop cum Writeshop 1. Tourism Structure Development Planning and Writeshop
Researchers: PSU (Arnel Zarcedo, Study Leader)

This Writeshop will develop the tourism structure plan of the MPLS TMP. The output of this Writeshop will serve as the first chapter of the TMP.

Key findings from various studies generated in this project will be used as inputs to this plan. Key participants will be selected to contribute and provide insights into the draft tourism structure.

Workshop cum Writeshop 2. Tourism Product Development and Promotion Planning and Writeshop
Researchers: PSU (Karlo Pedro Medroso, Study Leader)

This Writeshop will develop the tourism products, tourist site and tourism development area themes, as well as product positioning strategy and brand of the MPLS TMP. The output of this Writeshop will serve as the second chapter of the TMP.

Key findings from various studies generated in this project will be used as inputs to this plan. Key participants from DTI, DOT, PDA, LGUs, Tourism industry and academe will be selected to contribute and provide insights into the draft tourism products and themes.

Workshop cum Writeshop 3. Tourism Capacity Development Planning and Writeshop
Researchers: PSU (Patricia Candelaria, Stusdy Leader; Menchie del Castillo, Member)

This writeshop will develop the various skills that will enable each stakeholder in the area to capture the benefits of tourism of the MPLS as well as maximize their potential to contribute to the full development of sustainable ecotourism industry in the area. The output of this Writeshop will serve as the third chapter of the TMP.

Key findings from various studies generated in this project will be used as inputs to this plan. Key participants such as representatives from LGU, DOT, DENR, PDA,TESDA, DEPED and academe will be selected to contribute and provide insights into the draft tourism products and themes.

Workshop cum Writeshop 4. Tourism Livelihood Development Planning and Writeshop
Researchers: PSU (Margarita Lanciso, Study Leader; Irene Daet, Member)

This Writeshop will develop the various livelihood and rural enterprise that will enable each stakeholder in the area to capture the benefits of tourism of the MPLS as well as maximize their potential to contribute to the full development of sustainable ecotourism industry in the area. The output of this Writeshop will serve as the fourth chapter of the TMP.

Key findings from various studies generated in this project will be used as inputs to this plan. Key participants such as representatives from DTI, DA-BFAR, ATI, LGU, DOT, DENR, PDA,DOLE and academe will be selected to contribute and provide insights into the draft tourism products and themes.

Workshop cum Writeshop 5. Tourism User Fee and Sustainable Financing Planning and Writeshop
Researchers: PSU (Atty Jude Villar, Study Leader; Charlie Balagtas, Member)

This Writeshop will develop the user fee system to develop a sustainable tourism and resource management financing mechanism in the area to incentivize stakeholders and clienteles behavior for

the promotion of sustainable coastal resource ecosystem and tourism governance. The output of this Writeshop will serve as the fourth chapter of the TMP.

Key findings from various studies generated in this project will be used as inputs to this plan. Key participants such as representatives from LGU, DENR, PDA, tour operators and academe will be selected to contribute and provide insights into the draft tourism products and themes.

Workshop cum Writeshop 6. Tourism Carrying Capacity and Monitoring Planning and Writeshop
Researchers: PSU (Arnold Sabuco, Study Leader; RA Dioneda Jr, Member)

This Writeshop will develop the parameters, indicators and benchmarks for monitoring, the composition of monitoring teams, the process for generating data, developing mechanisms to implement compliance and communicating these mechanisms to stakeholders and clienteles in order to achieve compliance and promote sustainability of the tourism assets. The output of this Writeshop will serve as the fifth chapter of the TMP.

Key findings from various studies generated in this project will be used as inputs to this plan. Key participants such as representatives from LGU, DENR, PDA, tour operators and academe will be selected to contribute and provide insights into the draft tourism products and themes.

Workshop cum Writeshop 7. Integration of Multiple Planning Outputs into MPLS Tourism Master Plan
Researchers: PSU (Raul G Bradecina, Study Leader; Josenia Penino, Member)

This Writeshop will integrate all outputs of the Writeshop cum workshops for development planning to generate the draft of the MPLS TMP. This will be participated by the research project team leaders and the consultants.

## IV. References


Alcala, A. C. (1998). Natural Fishing Experiments in Marine Reserves 1983-1993: Community and Trophic Responses. Retrieved from Springer Link: http://link.springer.com/article/10.1007/s003380050144

Alcala, A.C., Gomez, E.D., 1979. Recolonization and growth of hermatypic corals in dynamite blasted coral reefs in the Central Visayas, Philippines. In: Proceedings of the International Symposium on Biogeography and Evolution in the Southern Hemisphere, Auckland, NZ, pp. 645–661.

Alcala, A.C. and C.R. Russ. 2002. Status of Philippine Coral Reef Fisheries Asian Fisheries Science 15 (2002):177-192 Asian Fisheries Society, Manila, Philippines.

Calumpong, H., Sienes, P. M., Santos, T. R., Padin, J. I., & Sabater, E. (2013). Plankton Abundance in Ticao Pass, Masbate, Philippines. Ecol.Env. and Cons., 47-51. Retrieved from Research Gate

Department of Environment and Natural Resources. (1990). Revised Water Usage and Classification/Water Quality Criteria Amending Section Nos. 68 and 69, Chapter III of the 1978 NPCC Rules and Regulations. DENR Administrative Order No. 34. Philippines

English, S, C Wilkinson and V Baker (1997). Survey Manual for Tropical Marine Resources. Townsville, Australia, Australian Institute of Marine Science, Townsville Australia: pp. 378

Establishment of Bongsanglay Natural Park. Proclamation No. 319 series of 2000.

Hutchings P. 1978. Non-colonial cryptofauna, p. 251-261. In D.R. Stoddart & R.E. Johannes (eds.). Coral reefs research methods, UNESCO, Monograph 19, Paris

Hutchings P. 1986. Biological destruction of coral reefs. Coral Reefs 4: 239-252.

Kohler KE, Gill SM (2006) Coral Point Count with Excel extensions (CPCe): a visual basic program for the determination of coral and substrate coverage using random point count methodology. Comput. Geosci. 32:1259–1269

## I. Basic Information

| | |
|---|---|
| Research Program Title: | Assessment, Total Economic Valuation of Coastal Habitats and Development of Sustainable Ecotourism Master Plan for Malabungot Protected Landscape and Seascape |
| Implementing Agencies: | Dept. of Environment & Natural Resources ROV (DENR-V), Legazpi City<br>Partido State University-Coastal Wetland Center (PSU-CWC), Sangay, Camarines Sur<br>Bureau of Fisheries and Aquatic Resources V (BFAR V), Bula, Camarines Sur<br>Bicol University-Center for Policy Studies and Development (BUCPSD, Legazpi City<br>Local Government of Garchitorena, Province of Camarines Sur |
| Key Proponent: | Ronnel A. Dioneda Jr.<br>Faculty-Researcher, PSU-CWC, PSU Sangay Campus<br>CP# 09272470226 email add: rondioneda@gmail.com |
| Duration: | August 2016 to January 2017 |
| Funding Requirement: | **PhP 1,500,000.00**<br>P1.3M from DENR-V and P200K from PSU-CWC |
| Funding Source | DENR RO V under its Coastal and Marine Ecosystem Management Program (CMEMP)<br>Partido State University-CWC |

## II. Rationale

The protected area system has long been practiced in the country primarily as part of the multitude of interventions to preserve natural endowments and to put in check degradations due to natural and anthropogenic activities. The Philippine Commission Act No. 648 (passed by the US congress in 1903) marked the pioneering foundation of the national parks and protected area initiatives in the country. The scope of these protected areas was expanded by Act No. 1148, also known as Forest Act in 1904. This later on gave birth to establishment of national parks by virtue of the enactment of the National Parks Act (Act No. 3195) in 1932. This was followed by the revised Forestry Code in 1975 and the National Integrated Protected Area System Act (RA 7586 of 1992). The later provides the legal framework for the establishment and management of protected areas in the Philippines. The NIPAS Act is part of the actualization of the commitment of the Philippine government in various international conventions and agreements it entered into. Under the NIPAS, protected areas in the country are classified and managed in varied ways. One of them are natural parks. NIPAS describe protected landscape/seascapes as areas of national significance which is characterized by the harmonious interaction of man and land (and the seas) while providing opportunities for public enjoyment through recreation and tourism within the normal lifestyle and economic activity of the area. Pursuant to NIPAS, the Malabungot Protected Landscape and Seascape (MPLS) was established in on April 23, 2000 under Proclamation No. 288, which was later on formally legislated under the senate bill no. 2895. The protected area is located at Malabungot Island of the municipality of Garchitorena and encompasses an area of and formally 120.62 hectares. There is no available baseline records that were used in establishing the area as NIPAS protected area nor any recent technical information showing monitoring. Around eight kilometers Northeast of Chico Island is a network of smaller MPAs established by Garchitorena LGU two years ago.

The sustainable utilization of critical ecosystems and habitats which provide the *reason d' etre* of seascape and landscape conservation enabling acts is commonly challenged by keeping a balance with its resource and social components. To achieve sustainability the management of these triad components as systems becomes imperative. Deeper understanding of the status of these systems and their dynamic interactions are key requirements for their rational management in the context of protected seascapes and landscapes.

For instance, corals, seagrass and seaweeds and mangroves are considered life support systems in the coastal zones. These aquatic habitats are production systems that produce fishery resources. However, anthropogenic activities and natural disturbances create a synergy of impacts that cast a cloud of uncertainty to their long-term existence and the provision of benefits from these habitats. Alcala (1998) demonstrated that a healthy reef system could provide as much as 18 mt/km$^2$/year while those in poor state could measly just provide 2 mt/km2/year. Considering the scarcity of technical baseline information for MPLS, there is a dire need to comprehensively assess its coastal ecology. The management of the MPLS and the attached resources should be well-supported with scientific information.

Coastal ecosystems and habitats, aside from the resources they produce, also provide tremendous benefits in forms of ecological services. The very strong reliance of people from marine ecosystems' benefits erodes both the quality and quantity of the services provided by these ecosystems. This is in the face of fast degradation of many coastal zones by the combined impacts of natural and intense man-made disturbances that are further compounded by climate change impacts. Commonly, these ecological services don't have monetary values as they don't have markets. The absence of monetary values imputed by society for these services results to inadequate investments of public funds by decision makers to promote their ecological integrity and health. Scholars recognize the fact that the degree at which these ecosystems effectively function and provide goods and services are dependent on their integrity and health (Calumpong et al., 2013). This underscores the need for the economic valuation of the services given by these ecosystems to generate the compelling basis for investing public funds for their sustainable management. Recent local studies found out that the economic value of goods and services from coastal resources in the five SCREMP sites in Albay is tremendous. Torres et al (2015) estimated annual foregone benefits could run to PhP 238 millions of pesos if the present poor too fair conditions of its coastal habitats prevail.

Over years, it has been observed that communities in fishing villages are the poorest. Social scientists attribute this observation to the open access nature of the resource as well as the lack of opportunities brought by the geographical remoteness of most fishing villages to economic hubs. The open access nature of the coastal resources results to the occurrence of the seminal tragedy of the commons in which common resource owned by everybody brings ruin to all. Also, the lack of opportunities in coastal fishing villages results to resource dependency, unsustainable resource use and economic vulnerability. While the establishment of institutions such as MPAs and enabling acts that declare coastal and marine areas as protected seascapes and landscapes served as management tools to address the tragedy of the commons, the challenge for developing sustainable financing mechanisms to incentivize resource users and stakeholders to support and promote sustainable resource governance weakens their effectiveness. This provides the imperative for establishing a payment for environmental services system in the form of users' fees. Whereas financial assistance and livelihood supports have been promoted to fishing communities in the past to uplift their living conditions, their being resource-based and extractive nature compounded by market accessibility constraints had casted cloud of doubts on their viability and sustainability.

Recently, sustainable ecotourism has been increasingly recognized to have the potential to address the problem brought by lack of opportunity in many fishing communities brought by its geographical remoteness. Living up to the mantra of "leaving nothing, except footprints; taking nothing, except memories", this innovative form of social enterprise aptly harmonizes with remote fishing communities as non-extractive livelihood strategy because of its ability to use tourism commons as entrepreneurial commodities where the market comes to the product. A marriage between coastal resource management and sustainable ecotourism will address the twin problems

of open access nature and lack of opportunities in fishing communities that fuels poverty and resource and ecological degradation. Developing a framework and a roadmap that comprehensively integrates coastal resource and biodiversity conservation and sustainable ecotourism that are both scientific information and market-driven will not only make MPLS as a tool for protecting biodiversity and rationally managing coastal resources, but also as an engine of growth that will fuel the rural economies of the communities surrounding it.

This research program would be a pioneering intervention in the MPLS to achieve the envisioned directions. In general, it would look into the present condition of the bio-ecological, resources and socio-economic dimensions attached to the resource utilization and biodiversity conservation as valuable inputs for the rational management and establishing a roadmap for sustainable ecotourism in the MPLS and adjoining area. Specifically, the research program intends to accomplish the following:

1) Investigate the diversity, community structure, density, percent cover, frequency cover, biomass of seagrass and seaweeds of MPLS together with accounts of observable wastage;
2) Assess the health of the coral reef systems in terms of percentage cover of major lifeforms and provide accounts of observable disturbances
3) Determine the diversity, species density, basal area and other measures of community structure for mangroves and provide accounts of observable perturbations.
4) Characterize the water quality and the 24-hour plankton dynamics of MPLS;
5) Deploy autonomous reef monitoring structure at MPLS and LGU MPA;
6) Characterize the capture fisheries and the reef associated reef fishes of Garchitorena with emphasis at MPLS;
7) Generate basic bathymetric map and coastal habitat maps of MPLS with estimate of areas per habitat;
8) Generate capacity development information needs of personnel in the existing industries in the MPLS area
9) Produce the socio-demographic information, perceptions and attitude towards coastal resource management and ecotourism development, tenure, resettlement and livelihood
10) Identify the resources and habitats available in the area, the community interventions to the resources and habitats, the problems/threats related with the resources and habitats
11) Generate the monetary value imputed by society for the non-use value of the MPLS seascape
12) Provide inventory of various tourism attractions covering both natural and cultural assets within the MPLS area.
13) Generate recreational attitude, preference, travel cost profile, and market segment profile
14) Develop potential livelihood program that will provide potential development of sustainable ecotourism industry in the area
15) Identify the factors affecting the compliance and sustainability of the tourism assets in the area.
16) Integrate all outputs of the Writeshops cum workshops for development planning to generate the draft of the MPLS TMP.

## III. Research Components, Methodologies and Expected outputs.

### Project 1. Coastal Ecological Assessment of MPLS (Marjorie Atole and XJ Dioneda)

This project will provide the much needed updated information regarding the status of the existing coastal habitats and the coastal environmental conditions of the protected area. The former would investigate the status of coral reefs, seagrass and seaweed beds and mangroves, using conventional ecological measures. Coastal environmental profile would look into the water quality and the plankton community it supports. Cryptobiota would also be investigated in this project using the BU-DENR-made Autonomous Reef Monitoring Structure (ARMS). Locations and extent of occurrence of the coastal habitats will be initially determined from existing records and inputs from the LGU of Garchitorena.

Study 1. Seagrass and Seaweeds (SGSW) assessment of MPLS
Researchers: PSU (X.J. Dioneda, Study Leader; BFAR Miles, and Christian)

Basic initial profiling and mapping of the location and distribution of seagrass and seaweed communities in each municipality shall be carried through reconnaissance surveys. Informal consultation with local folks shall be undertaken to ascertain and finalize the location of sampling sites. Number and location of stations shall be determined after the preliminary field survey.

Comprehensive S G S W assessment shall be undertaken using the conventional line quadrat method (LQM, English et al 1997). Minimum of three sampling stations will be assessed within the natural park. Each station shall have three 50-meters transects as replicates. Laid perpendicular to the shoreline, these three replicate transects will be laid continuously seaward whenever possible (i.e., extent of seagrass community presence seaward is 150 meters or longer). A 0.5x0.5 meter quadrats will be laid every 10 meter distance. Seagrass and seaweeds that are covered by the quadrats shall be assessed based on species presence, frequency cover and percent cover by species. If allowed for scientific research purposes, SGSW would be harvested for comprehensive assessment of species biomass estimation. The following shall be established in this study

a. Seagrass and seaweeds species observed in the sampling stations
b. SGSW species density, frequency, percent cover, canopy height for Thalasia sp.
c. Biomass per species (when sample harvesting is allowed)
d. Community structure of the SGSW measured in conventional ecological indices.
d. Accounts of the associated flora and fauna, disturbances and existing resource uses.
e. Establishment of at least six permanent monitoring stations which are geo-referenced and marked with permanent markers for long-term SG/SW monitoring.

Study 2. Coral reef Assessment of MPLS
Researchers: PSU: (R.A. Dioneda Jr. Study Leader; Noel Ombao, Member, Joel Balasta, and Garner)

Coral reefs have been traditionally protected through the marine protected area system of many local government units (LGUs). Many MPAs in the Philippines were established based on the presence and the intended protection of this coastal habitat. This can be due to the documented economic and ecological benefits they provide. In this study, the coral reef systems of Malabungot MPLS would be subjected to comprehensive assessment.

Location of the coral reef areas shall be initially determined from existing records and coordinative discussions from concerned people in Garchitorena. Validation shall be done through reconnaissance survey (manta tow technique) coupled with the traditional knowledge of local fishers. This is to determine the relative location and concentration of the coral reef systems in each municipality and help identify sampling stations. At least three stations will be identified in Malabungot reef systems. Two methodologies will be carried-out in coral assessment. First is the conventional Line Intercept Transect (point intercept) outlined by English et al (1997). This will be reinforced by coral photo transect method developed by Roelfsema and Phinn (2009) of the university of Queensland. Both methods shall be used in all stations, aided with SCUBA diving

system and shall use 50 meters transect. LIT would be carried out by videotaping the lifeforms traversed by the transect line. Reading of point intercepts would be carried-out in the laboratory and analysis would be carried-out in excel. Photo transect would use the minimum required paraphernalia (GPS with PC interface capability, wide angled lensed underwater camera, plumb line, transect line, floats, dry bag and reel). Photos will be analyzed using standardized software Coral Point Count excel (Kohler and Gill, 2006) or other simpler alternative analytical softwares.

The same set of assessment protocols would be employed in assessing the status of the coral reef systems of the three MPAs established by the LGU in Garchitorena. Although outside of the geographical coverage of the MPSL, providing technical information about the reef systems of the LGU-established MPAs would substantiate the existence of these MPAs. This is on the light of the fact that their establishment was devoid of any comprehensive technical basis (i.e., the MPAs are put in place just on the basis of coral occurrence from manta tow surveys done three years ago).

The following are the deliverables under this study:

a. Percent cover estimates of coral substrate types and benthic organisms and their criteria ratings outlined by Gomez and Alcala (1979);
b. Documentation of possible reef impacts such as cases of bleaching, poison and blast fishing, diseases, trashes, anchorage, fishing and other perturbations;
c. Geo-referenced sampling sites. Transects from selected stations shall be designated as permanent monitoring stations marked with monitoring blocks with the prescribed dimensions;
d. Geo-tagged photos of the assessment conducted; and,
e. Comprehensive report for the coral reef assessment and fish visual census

Study 3. Assessment of Mangroves at MPSL

Researchers: PSU (X.J.Dioneda: Study Leader; Members: May Vinas and Jasper Nieves)

As seen in Google earth, mangroves of MPLS appear to just occur at the southern, eastern and northern sections of Malabungot Island. Preliminary site inspection is needed to ascertain their occurrence and to be able to design appropriate sampling design. Line Plot method (English et al 1997) shall be employed to assess the mangrove communities of MPLS. At least three transects (100 meters when possible) shall be laid per mangrove cluster (i.e., southern, eastern and northern). Coordinates of stations shall be obtained and appropriate marking will be obtained for establishment of permanent monitoring stations. The following shall be obtained in this study:

a. Species and count of trees, saplings and seedlings per species (regenerative capacity)
b. Percent crown cover
c. Basal area per species and basic ecological indices of the mangrove communities per section/cluster.
d. Documentation of observable perturbation
e. Geo-tagged photos of the assessment conducted; and,
f. Comprehensive report for the mangrove assessment.

Study 4. Water quality and the plankton community of MPLS waters

Researchers :(RA Dioneda Sr. Study Leader; R.A. Dioneda, Jasper Nieves and Sci. Res. Aide)

Plankton community dynamics is strongly influenced by the day and night cycle. Species composition and abundance change at certain location and depth as plankton tend to be strongly respond to and affected by light and dark cycle and by current driven by tide and other forces. A 24-hour sampling of plankton on key station would then help describe in full detail the bio-ecological dynamics of plankton of the natural park. This study would then intend to establish the following: 1) Plankton diversity and abundance in the natural park 2) Temporal plankton profile within a 24-hour cycle 3) Document vertical migration of zooplankton in a specific station. These objectives will be achieved through a 24-hour plankton sampling. Water sample of known volume at certain depth will be collected using a Kemmerer water sampler. Samples will be taken from the bottom and with a one meter interval thereafter. Samples will be immediately filtered at 60 um (for

zooplankton) and at 20um for phytoplankton. Concentrated samples will be preserved using acid lugols solution and microscopy will be held at the PSU-CWC laboratory.

Together with the 24-hour plankton sampling is the determination of the basic physical and chemical properties of the water of a certain station in the natural park. Portable multiparameter meters will be used here to measure pH, DO, TDS, salinity, temperature and other parameters. Data loggers (Hobo) to record the temperature and light intensity will be deployed for at least two days at the bottom and near-surface locations. Light penetration will be measured using secchi disk. Some water samples will be collected and fixed for nutrient analysis (nitrate, phosphate). Water quality measures shall be compared to the DAO 2016-08 standards.

Study 5. Deployment of Cryptobiota Settlement Modules at MPLS.
Researchers: (RA Dioneda Jr, Study Leader; With assistance of other divers)

Reefs are dynamic systems which continually undergo active interplay between destructive bioerosion and constructive formation (Schuhmacher 1978). The speed at which a reef is formed depends on rates of skeletongenesis and consolidation, while destruction rates depend upon physical, chemical and biological eroding processes (Risk and Sammarco 1982, Hutchings 1986). The role of cryptic organisms on both processes have been documented to be crucial. They increase the complexity of substrates which cons-antly create spaces for coral larvae and other organisms, thereby maintaining diversity (Hutchings 1986). Taken with the degrading impacts of factors driven by climate change, fate of coral reefs maybe driven more on a destructive end. This research adopts the definition cryptic fauna (or cryptofauna or cryptobiota) of Hutchings (1978), Peyrot-Clausade (1981) and Scott (1987) which means organisms living in small pores, holes or cavities within the rock or inside organisms that cover them.

The purpose of this project is to describe the diversity ad abundance of cryptic fauna associated with coral reefs. This will also provide opportunities to test the locally designed cryptobiota settlement trap (CST). As cryptic organisms, they are less studied and are rarely investigated in conventional ecological studies. But considering the above-stated ecological importance, investigation of cryptobiota would be necessary especially to protected areas that are expected to repopulate unprotected zones by spillover or by other means.

Following the autonomous reef monitoring structure (ARMS) schematics of the Smithsonian Institute, Bicol University and DENR-V fabricated ARMS modules made of acrylic plastic. Two ARMS modules will be deployed in the reef systems of MPS and in the LGU MAP. These ARMS would be coupled with temperature data logger and would be maintained in geo-referenced locations for a year. SCUBA-aided photo documentation would be made at least two times within the project duration. Likewise, data from the logger would be downloaded and the logger would be redeployed for long-term data logging. Metagenomic studies will be employed in the future to reveal the cryptobiota assemblage of MPSL reef systems.

**Project 2. Capture Fisheries and Reef Associated Fishes of MPLS (Redentor Buitre and RA Dioneda Sr)**

Study 6. Capture fisheries of Garchitorena with emphasis on Malabungot waters.
Researchers (Redentor Buetre, Study Leader and Irene Daet, Member)

Fisheries is one of the primary extractive economic activity that impacts coastal resources' long term abundance. Its potential to decimate a fish stock is very high especially when the fishing intensity is beyond the regenerative capacity of the fish. There is then a need to characterize the fishery resource extraction in Malabungot and adjoining areas. One way to do this is to characterize the capture fisheries of the area. Along this line, catch characterization would be done by direct catch and landing observation/surveys, market surveys and recall interview with key informants. Seasonality of fishing operation, resource use conflicts/issues and other fisheries concerns would be determined by focused group discussions and through semi-structured interview. Type and number of fishing gear

implements shall be obtained by gear inventory and using the Fish and boat R data of BFAR and LGU records.

Study 7. Reef Fish Assemblage and Biomass of MPSL and Garchitorena LGU MPAs.
Researchers: PSU: (R.A. Dioneda Study Leader; BFAR: Janlops and Joel Balasta)

The variety, abundance and biomass of the coral reef fishes are directly attributed to health of reef systems. Good reef fish assemblage also support tremendous supply of fish catch reaching to about 25% of coastal fisheries productivity (Alcala and Russ 2002). There is then a need for the inclusion of the reef fish monitoring in CRM monitoring activities. Along this line, fish visual census (FVC) will be carried-out in selected stations used for coral reef assessment. Here, a diver would identify count and estimate the approximate total lengths of the reef fishes encountered along the entire stretch of the transect line used for coral assessment. This study would establish the following: 1) diversity of reef fishes in station, 2) Biomass per species and total biomass for the station, 3) notes on indicator species and 4) community structure defined by conventional ecological indices per station.

**Project 3. Coastal Habitat and MPLS Location Revalidation Mapping and Basic Bathymetry of MPSL** (RA Dioneda, Jr and Mapping expert)

Study 8. Mapping of coastal habitats of MPLS.
Researchers: DENR

Distribution and expanse of the coastal habitats shall be determined using GPS-based survey around Malabungot Island. Boat sorties equipped with GPS with PC interfacing capabilities shall be made along the edges of the reef systems, sea grass and seaweeds and the sea-ward side of mangroves. Trek by foot shall be one along the land-ward edge of the mangrove areas. These geo-referenced information shall be used for digitized mapping process using acceptable tools (GIS or manifold). This study intends to come-up with a map the coastal habitats of the MPSL with estimated of area per habitat.

Study 9. Revalidation Survey of Location and Basic Bathymetry of MPLS
Researchers: PSU (R.A. Dioneda) and DENR

The coordinates of both the Proclamation 288 and the Senate Bill No. 2895 yielded erroneous coordinates for the point 1 reference. When plotted, it's near Atulayan Island of Sangay, Camarines Sur. If the bearings and defined distances between points would be traced, it would draw a polygon inside Lagonoy Gulf, around a kilometer to the East of Atulayan Island. Being a protected area with access restrictions and regulations, problematic technical definitions of its locations would have wide ranging implications to law enforcement and management decisions. There is then a need for a revalidation study to correct this technical flaw.

To characterize the bottom topography of the immediate water of Malabungot Island, a bathymetric survey will be carried-out. A Garmin Fishfinder Striker 4 will be used to characterize bottom contour around the island. Additional GPS (Garmin 7160) units will be used to navigate preset routes. Basic depth profile (in defined isobaths if possible) around the island would be the basic output of this activity.

**Project 4. Socio-Economic Assessment (Patricia Candelaria and Luisa Lanciso)**

Study 10. Training Needs Assessment
Researchers: PSU (Patricia Candelaria, Study Leader; Menchi del Castillo, Member)

The training needs assessment survey will generate the information of the capacity development needs of personnel in the existing industries in the MPLS area. The survey will generate the following data:

Priority capacity development needs in trade skills

Priority capacity development needs in food product and development skills
Priority capacity development needs in agriculture livelihood development skills
Priority capacity development needs in fisheries livelihood skills
Priority capacity development needs in enterprise development
Priority capacity development needs in tourism management skills
Priority capacity development needs in tourism enterprise development skills
Priority capacity development needs in upland and coastal resource management skills

It will employ key informant interview as data generating tool and using key stakeholders in the industry sector as well as the LGU and NGA as key informants. The industry stakeholders key informants will include owners of existing resorts, inns, hotels, small scale, medium and large scale commercial and service-oriented establishments in the area such as parlors, barber shops, transport, repair shops, food processors and producers, tour operators, sari-sari stores and groceries.. The key informants in the LGU, and NGAs will include selected officials of the barangay, DEPED, and other NGA offices available in the area.

The information generated will be used to complement and enhance the TNA data that will be captured through household survey questionnaire. These data will serve as inputs in developing the capacity development plan of the MPLS Tourism Master Plan.

Study 11. Socioeconomic Assessment
Researchers: PSU (Irene Daet, Study Leader)

The socioeconomic assessment will generate the socio-demographic information, perceptions and attitude towards coastal resource management and ecotourism development, tenure, resettlement and livelihood. It will be implemented through a household survey using questionnaire as data generating tool.

The first part of the questionnaire will include personal information of the household respondent such as age, sex, educational attainment, number of children, years in fishing, years in farming, occupation (major, secondary and tertiary), fishing income, income from major, secondary, tertiary occupation, fishing operation expense,

The second part of the questionnaire will include household respondents' knowledge attitude, practices and perceptions. The knowledge part will include awareness on NIPAS-MPLS, attendance to trainings, awareness on the trainings conducted by NGAs, awareness of fish sanctuaries and reserves, level of awareness of MFRS, awareness of environmental laws, ordinances implemented, awareness of the effectiveness of environmental laws implemented, awareness of the current status of MPLS, factors affecting marine resources and awareness of the effect of climate change on coastal ecosystem and resources. The attitude part will include opinion on the current status of marine resources in the MPLS, measures to sustain the coastal and marine resources in the MPLS, PAMB to manage the MPLS, agency that would spearhead the protection and management activities of the MPLS. The practices shall include membership to organization, involvement in activities being done on MPLS, fishing routine, income from fishing, creation of biodiversity friendly livelihood activities, and identified livelihood activities. The perception will include perception on the quality of the coastal and marine ecosystems and habitats as well as the status of the resources 10 years from now and at present. It shall also capture their perception of the threats in terms of their scope, intensity and area covered to attribute such status. The questionnaire will adopt the insights and format of DENR knowledge attitude and practices questionnaire.

The third part of the questionnaire will include items probing on their attitude towards the coastal ecosystem and its resources as well as the directions for sustainable utilization of resources and protection of the coastal and marine ecosystems. It shall also capture their attitude towards sustainable tourism promotion and governance.

The fourth part of the questionnaire will probe on the information of their dwellings that will include, number of years residing in the area, birth place, residence of origin, status whether tenured or non-tenured migrants, size of the dwellings or lot area occupied, farm area occupied, nature of dwelling whether native, semi concrete and concrete, whether willing or unwilling to be resettled, reasons for willingness or unwillingness to resettled, perception of risk to dwelling.

The fifth part shall probe on the materials style of life of the household. This will include enumeration of the available modern day indicators of living standards such as availability of comfort rooms, power, piped water, appliances, service vehicles, communication gadgets, etc.

The sixth part will capture the current livelihood engagement of the household respondents, the available livelihood assets in the households that could be categorized under human (skills such as carpentry, baking, food processing, sculpting, etc), production assets (fishing and farming assets such as carabao, thresher, boats, nets, engine), farms, buildings, water supply, fishponds, equipment etc..

The survey uses the household as the unit of analysis. A two stage randomized probability sampling design will be adopted and the number of sample respondents will be estimated following Slovin's sample formula. In this design, the universe will be divided into rural, urban and island sites. The first stage sampling will randomly select one half of the communities in these sites. In the second stage sampling randomization, household respondents from the randomly selected communities in each site category will be randomly selected. Randomization will use the electronic record of the household head names from the data bank of COMELEC or the LGU and shall be selected through the randomization feature of Microsoft EXCEL Spreadsheet.

Study 12. Resources and Resource Use Patterns
Researchers: PSU (Luisa Lanciso, Study Leader, Member)

A PRA will generate the resources and habitats available in the area, the livelihood attached to the resources and habitats, the problems/threats related with the resources and habitats. It shall also generate seasonality of fishing, the current status of the resources and the ecosystem, the interventions done for CRM, the institutions involved, the activities of male and female members of the households.

The FGD will generate data on household, reproductive health, migration and occupation data, community projects and organization and stakeholders, environment-friendly enterprises, community and fishery organizations, influential stakeholders, community perceptions on the state of natural resources in terms of mangrove, coral reefs, seagrass, beaches, estuaries and upland forest, fishery coastal activities, number of fishers, fishing gears, boats and catch rates, coastal resource management initiatives and environmental awareness, name of marine habitats or organisms that are sensitive or vulnerable, list of coastal resources that were present before 10 years or more, fishery resources condition, most important resources (top ten fish species, crustaceans and mollusks), perceived changes in tp ten fishes, crustaceans and mullosks in the last ten years (before and after)

Two sites will be selected for the PRA and FGD, one in the island (Malabungot Island) and one in the fishing community (Bgy Binagasbasan). At least 15 key informants will be selected from representative groups of women, youth, barangay officials, elderly, NGO, academe, fishers, farmers, resort owners, private sector etc.

The PRA will employ Venn diagram, Resource Map, Gear calendar diagram, species calendar diagram, diagram for habitat and 24-hour activity, trend diagram for fish catch. The FGD will employ KI questionnaire adopting RARE PRA and FGD protocol.

**Project 5. Economic Valuation of Environmental Services and Establishment of User Fee System (Raul G Bradecina and Leih Ann Odono)**

Study 13. WTP for Protecting MPL Seascape and MPA Survey
Researchers: PSU (Lieh Anne Odono, Study Leader; Raul G Bradecina, Member)

The study will generate the monetary value imputed by society for the non-use value of the MPLS seascape. It will employ the contingent valuation method. A preliminary survey will be done to determine the minimum and maximum bids imputed by municipal resident and fishers for the protection of MPLS ecosystem. Final bids will be determined and a questionnaire survey will be developed using single bounded dichotomous choice bid elicitation approach. Respondents will be selected following randomization using Slovin formula. A two stage probability sampling will be done to select the estimated number of respondents. The first stage will identify randomly the urban

and rural community sites from the total list of the barangays in the municipality. The second stage will identify the respondents in the randomly selected sites. Data will be analyzed using NLOGIT software to determine MWTP and the parameter estimates of the WTP curve.

Study 14. Establishing User Fee System
Researchers: PSU (Atty. Jude Villar, Study Leader)

This study will determine the acceptability of establishing user fee system, identify what mechanism should be established to collect the user fee, the collecting entity and how the process by which the collected fees shall be disbursed or allocated. These data will be generated through survey and workshops with stakeholders. The question items for the data will be integrated in the WTP survey. In addition, the data will also be captured through writeshop using FGD. The participants in the FGD will be selected from representative groups of resort owners, shop owners, LGU officials, NGO, NGAs, tourists, fishers and farmers, academe.

Study 15. Total Economic Valuation of MPLS
Researchers: PSU (Raul G Bradecina, Study Leader)

This study will estimate the total monetary value of MPLS using the total economic valuation concept. This will include the estimation of the use value and the non-use value of the MPLS. The use value will focus on the resource of the coastal and marine ecosystems with markets. This will include the recreational value, as well as the economic production of the habitats captured through fishing and similar extraction economic activities. The non-use value will focus on the environmental services of the MPLS that don't have market. This will include the value of protecting MPAs and seascape for future use.

The study will use the data generated from capture fisheries surveys and FISH R data bank to analyze and estimate the use-value of the resource. It shall also impute the recreational value of the resource generated from the recreational tourism attitude and preference survey focusing on the travel costs given by respondents. The non-use value of environmental services of MPLS will be captured using CVM. This will generate the mean and aggregated WTP for protecting the MPA and seascapes.

**Project 6. Ecotourism Development Research (Charlie Balagtas and Josenia Penino)**

Study 16. Tourism Attraction Inventory
Researchers: PSU (Josenia Penino, Study Leader)

This study will generate the various tourism attractions covering both natural and cultural assets within the MPLS area. These attractions will be identified within each barangay within the MPLS area. Document analysis, key informant interviews and FGD will be held to elicit attractions with ecotourism potential and provide brief description on these tourism assets. The document analysis will involve several review activities of the Municipal Development Plan and Profile, Previous Studies and Comprehensive land Use Plans, Municipal Tourism Development Plan to determine existing and potential tourism attractions and sites. The key informant interview will also inquire from residents the areas and assets in their localities that have tourism promise. The FGD will also request the participation of municipal officials and fisher representatives to identify and locate key attractions in the MPLS that are only known to them.

Study 17. Recreational Attitude, Preference and Travel Cost Study
Researchers: PSU (Charlie Balagtas, Study Leader; Raul G Bradecina, Member)

This study will generate the tourist attitude and preferences, the frequency of visit, potential expenditures that will be incurred from the travel. It wil also generate the recreational value of the MPLS tourism and the determinants of tourists' willingness to visit MPLS. This will be generated through a survey using a questionnaire.

The first part of the questionnaire will include personal information, income and occupation and the residence of the respondents.

The second part of the questionnaire will include information on tourists travel attitude. This will cover the data on respondents willingness to travel to the place and the reasons for the decision to travel, the most likely months, weeks and days in a year he would like to travel, the frequency of travel, the expenses he would most likely spend, whether travelling alone, with companion, or family, the number of days he would like to stay, the type of transport and route preferred, the accommodation preferred and reasons for the answers.

The third part of the questionnaire will include information on tourists' preference on tourism destinations/attractions, experience and accommodation. These will include data on the preferred qualities of tourism destinations, experience and accommodation. Quality indicators for destinations, experience and accommodations will be generated and shall be assigned rank using scales of 1-5 to measure level of preference. The destinations will cover enumeration and selection options of various MPLS attractions including natural habitats such as coral reefs, seagrass, mangroves; including experiences such as boating, snorkeling, sand strolling, beach sports, game fishing, swimming, beach picnics, vacations, adventures; including cultural attractions such as community visits, barangay museums, local festivals, churches, life in the rural coastal communities, livelihood, agritour, and culinary.

The survey will involve a total of 300 respondents to be interviewed. The respondents will cover potential local tourists. Of the 300 respondents, 90 will be interviewed from tourists visiting Caramoan, Camarines Sur (to represent national tourists). The remaining 210 respondents will be distributed to respondents from selected towns of Camarines Sur namely Goa-Tigaon (70), Pili-Naga (70), and Iriga-Nabua (70). Enumerators will be hired and trained prior to the survey.

The data will be analyzed using descriptive statistics to characterize attitude and preferences of potential tourists. The recreational value of the MPLS and the factors that influence the tourist's decision to visit will be determined using zonal travel cost method following the routines of NLOGIT econometric software.

Study 18. Tourism Structure Development
Researchers: PSU (Arnel Zarcedo, Study Leader)

This study will develop in details the tourism structure of MPLS. It will identify the various tourism sites where tourism development areas will evolve by order of development priorities, the designated tourism development clusters, the themes to be developed for each cluster or tourism development area, the amenities required, and the structure of tourism circuits. This will be developed though a preliminary tourism structure development Writeshop. In this activity, the key findings in studies 17, 18 and 19 will serve as inputs. Other documents such as PDA Tourism Master Plan, Bicol Regional Development Plan, and the Garchitorena Municipal Development Plan and Profile will also be used as basis in the analysis. The tourism structure development framework prescribed by the DOT will be adopted during the Writeshop.

A workshop for finalizing the draft tourism structure will be conducted. In this activity, the draft tourism structure will be presented as working paper to the participants to facilitate discussion and development of outputs. The participants will review and revise the draft basing from their perspectives and insights as well as personal knowledge of development directions in their own areas of concerns. Participants to the workshop will be composed of tourism officers, municipal planning officers and committee chairman for tourism and infrastructures of legislative arms of LGUs, PDA, DENR, DPWH and DOT representatives, and academe.

Study 19. Tourism Carrying Capacity and Coastal & Marine Environment Quality Indicators Assessment
Researchers: PSU (Arnold Sabuco, Study Leader); Jett Atole, Member)

This study will generate the data for the carrying capacity of selected identified key tourism sites in the MPLS area as well as current status of key coastal and marine environment quality parameters such as pH, Turbidity, salinity, dissolved oxygen etc.

Carrying capacity will be generated using GIS due to time and resource constraints inherent in the project schedules. However, if time permits, "ground truthing" will be done for GIS-based parameters by field researchers. Constant values that are required in the equation for the estimation

of carrying capacity will be adopted from existing literature due to the fact that the information will be developed on an ex-ante basis that is before actual occurrence of tourism in the area.

Quality indicators will be generated through two-shot samples using multi-parameter kits to generate benchmark data. Other parameters will be used from the data generated during the ecological survey.

**Project 7. Sustainable Ecotourism Planning (Arnel Zarcedo and Raul G Bradecina)**

Workshop cum Writeshop 1. Tourism Structure Development Planning and Writeshop
Researchers: PSU (Arnel Zarcedo, Study Leader)

This Writeshop will develop the tourism structure plan of the MPLS TMP. The output of this Writeshop will serve as the first chapter of the TMP.

Key findings from various studies generated in this project will be used as inputs to this plan. Key participants will be selected to contribute and provide insights into the draft tourism structure.

Workshop cum Writeshop 2. Tourism Product Development and Promotion Planning and Writeshop
Researchers: PSU (Karlo Pedro Medroso, Study Leader)

This Writeshop will develop the tourism products, tourist site and tourism development area themes, as well as product positioning strategy and brand of the MPLS TMP. The output of this Writeshop will serve as the second chapter of the TMP.

Key findings from various studies generated in this project will be used as inputs to this plan. Key participants from DTI, DOT, PDA, LGUs, Tourism industry and academe will be selected to contribute and provide insights into the draft tourism products and themes.

Workshop cum Writeshop 3. Tourism Capacity Development Planning and Writeshop
Researchers: PSU (Patricia Candelaria, Stusdy Leader; Menchie del Castillo, Member)

This writeshop will develop the various skills that will enable each stakeholder in the area to capture the benefits of tourism of the MPLS as well as maximize their potential to contribute to the full development of sustainable ecotourism industry in the area. The output of this Writeshop will serve as the third chapter of the TMP.

Key findings from various studies generated in this project will be used as inputs to this plan. Key participants such as representatives from LGU, DOT, DENR, PDA,TESDA, DEPED and academe will be selected to contribute and provide insights into the draft tourism products and themes.

Workshop cum Writeshop 4. Tourism Livelihood Development Planning and Writeshop
Researchers: PSU (Margarita Lanciso, Study Leader; Irene Daet, Member)

This Writeshop will develop the various livelihood and rural enterprise that will enable each stakeholder in the area to capture the benefits of tourism of the MPLS as well as maximize their potential to contribute to the full development of sustainable ecotourism industry in the area. The output of this Writeshop will serve as the fourth chapter of the TMP.

Key findings from various studies generated in this project will be used as inputs to this plan. Key participants such as representatives from DTI, DA-BFAR, ATI, LGU, DOT, DENR, PDA,DOLE and academe will be selected to contribute and provide insights into the draft tourism products and themes.

Workshop cum Writeshop 5. Tourism User Fee and Sustainable Financing Planning and Writeshop
Researchers: PSU (Atty Jude Villar, Study Leader; Charlie Balagtas, Member)

This Writeshop will develop the user fee system to develop a sustainable tourism and resource management financing mechanism in the area to incentivize stakeholders and clienteles behavior for

the promotion of sustainable coastal resource ecosystem and tourism governance. The output of this Writeshop will serve as the fourth chapter of the TMP.

Key findings from various studies generated in this project will be used as inputs to this plan. Key participants such as representatives from LGU, DENR, PDA, tour operators and academe will be selected to contribute and provide insights into the draft tourism products and themes.

Workshop cum Writeshop 6. Tourism Carrying Capacity and Monitoring Planning and Writeshop
Researchers: PSU (Arnold Sabuco, Study Leader; RA Dioneda Jr, Member)

This Writeshop will develop the parameters, indicators and benchmarks for monitoring, the composition of monitoring teams, the process for generating data, developing mechanisms to implement compliance and communicating these mechanisms to stakeholders and clienteles in order to achieve compliance and promote sustainability of the tourism assets. The output of this Writeshop will serve as the fifth chapter of the TMP.

Key findings from various studies generated in this project will be used as inputs to this plan. Key participants such as representatives from LGU, DENR, PDA, tour operators and academe will be selected to contribute and provide insights into the draft tourism products and themes.

Workshop cum Writeshop 7. Integration of Multiple Planning Outputs into MPLS Tourism Master Plan
Researchers: PSU (Raul G Bradecina, Study Leader; Josenia Penino, Member)

This Writeshop will integrate all outputs of the Writeshop cum workshops for development planning to generate the draft of the MPLS TMP. This will be participated by the research project team leaders and the consultants.

## IV. References


Alcala, A. C. (1998). Natural Fishing Experiments in Marine Reserves 1983-1993: Community and Trophic Responses. Retrieved from Springer Link: http://link.springer.com/article/10.1007/s003380050144

Alcala, A.C., Gomez, E.D., 1979. Recolonization and growth of hermatypic corals in dynamite blasted coral reefs in the Central Visayas, Philippines. In: Proceedings of the International Symposium on Biogeography and Evolution in the Southern Hemisphere, Auckland, NZ, pp. 645–661.

Alcala, A.C. and C.R. Russ. 2002. Status of Philippine Coral Reef Fisheries Asian Fisheries Science 15 (2002):177-192 Asian Fisheries Society, Manila, Philippines.

Calumpong, H., Sienes, P. M., Santos, T. R., Padin, J. I., & Sabater, E. (2013). Plankton Abundance in Ticao Pass, Masbate, Philippines. Ecol.Env. and Cons., 47-51. Retrieved from Research Gate

Department of Environment and Natural Resources. (1990). Revised Water Usage and Classification/Water Quality Criteria Amending Section Nos. 68 and 69, Chapter III of the 1978 NPCC Rules and Regulations. DENR Administrative Order No. 34. Philippines

English, S, C Wilkinson and V Baker (1997). Survey Manual for Tropical Marine Resources. Townsville, Australia, Australian Institute of Marine Science, Townsville Australia: pp. 378

Establishment of Bongsanglay Natural Park. Proclamation No. 319 series of 2000.

Hutchings P. 1978. Non-colonial cryptofauna, p. 251-261. In D.R. Stoddart & R.E. Johannes (eds.). Coral reefs research methods, UNESCO, Monograph 19, Paris

Hutchings P. 1986. Biological destruction of coral reefs. Coral Reefs 4: 239-252.

Kohler KE, Gill SM (2006) Coral Point Count with Excel extensions (CPCe): a visual basic program for the determination of coral and substrate coverage using random point count methodology. Comput. Geosci. 32:1259–1269